Causal Optimizer Interaction Calculus: Hidden Geometric Relaxation and Identifiable Interventions

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Abstract

An optimizer experiment observes responses to algorithmic configurations without uniquely revealing hidden mechanisms. We develop a causal optimizer interaction calculus that separates pathwise realization, Möbius decomposition, and experimental identification. Under a fixed innovation coupling, every finite-horizon innovation-driven optimizer admits a canonical behaviorally minimal pathwise realization. Over any lower-finite intervention poset, its expected response has unique pure effects; for every finite support and design, an incidence operator gives the complete observational gauge, exact identifiability, sharp quotient stability, held-out predictions, and exact noiseless configuration complexity. The companion P1 paper proves that smooth hidden relaxation gives reduced-value curvature $-G^*H^{-1}G$ and that Boolean contrasts integrate it. Taking this as a structural input, we prove an observable-readout transfer theorem: arbitrary smooth update or trace readouts inherit an explicit five-term interaction through first and second hidden responses, with no universal sign beyond the reduced-value case. A third theorem develops Gaussian quotient minimax risk, exact confidence sets and tests, misspecification decomposition, certified decisions, and optimal replication. A controlled real-data experiment closes the reduced-value chain on a 65-dimensional strongly convex logistic model. Boolean effects and independent curvature integrals agree within 4.21×10^{-11}; nine unobserved continuous intensities agree within 8.88×10^{-13}; 100,000 Gaussian campaigns attain 0.9497 ellipsoid coverage and 0.9626 held-out power; and 4500 real-minibatch observations with 20,000 bootstrap draws reject order-two support with interval $[0.001839, 0.004399]$. Finite-response error decreases from 1.43×10^{-2} to 5.55×10^{-17} within its strong-convity bound. Neural trace audits provide complementary nonconvex response-class evidence. The paper contributes the causal, readout, and identifiable experimental layer while assigning the underlying Schur and path-space laws to P1 and P2.

1 Introduction

An optimizer is an intervention-dependent dynamical system. Its state may contain parameters, momentum, curvature estimates, preconditioners, schedules, clipping rules, random innovations, target transformations, and numerical state. A recorded parameter update is one projection of that extended dynamics. It generally admits many internal explanations: positive geometry may reproduce part of the update, while memory, operators, control, noise, or target changes reproduce the remainder.

This ambiguity creates two distinct scientific questions. A *response-class question* asks whether an observed trace is close to the outputs generated by a declared family, such as bounded diagonal geometry. A *mechanism question* asks how the counterfactual response changes when a declared module is altered. Passive projection can reject a response class. Causal mechanism effects require interventions, instrumentation, or an observation structure with equivalent identifying power.

We take the complete interventional response as the primitive object. A lower-finite poset $(\mathcal{P}, \preceq)$ indexes configurations, including Boolean sector masks, graded mechanism levels, and nested geometry budgets.

Under a protocol that fixes initialization, data, horizon, response map, and noise coupling, each configuration a produces a finite-horizon response $F(a)$ in a Hilbert space. The optimizer may be nonsmooth, stochastic, stateful, delayed, and history-dependent.

Three operations then have separate roles. Causal realization turns histories into a predictive state. Möbius inversion changes coordinates from configuration responses to pure interventional effects. A linear design operator maps those effects to queried observations. Their order is typed:

$$\text{causal dynamics} \longrightarrow \text{response map} \longrightarrow \text{Möbius effects} \longrightarrow \text{observational quotient.}$$

The universal theory requires no optimizer energy or geometry.

The companion P1 paper already answers the structural reduced-value question. When intervention amplitudes couple to hidden optimizer variables that relax by energy minimization, it proves that the reduced intervention Hessian is

$$D^2_{uu}\bar{E} = -G^* H^{-1} G,$$

and that Boolean reduced-value contrasts integrate this curvature (Li, 2026b). P2 supplies the corresponding path-space Möbius–Jacobi theorem (Li, 2026c). The open problem for an experimental calculus is that optimizer studies usually observe updates, traces, or task statistics rather than the reduced optimal value itself.

We solve this interface problem with an observable readout theorem. For any smooth response $F_R(u) = R(u, \sigma_\star(u))$, its mixed derivative is an explicit five-term transfer through the first and second hidden responses. The P1 negative Gram law is recovered for the reduced-value readout, while an actual update readout generally has no universal sign. Its Boolean Möbius effect remains the exact integral of this transferred curvature and is therefore accessible to a factorial design. This separates three statements that had previously been conflated: P1 explains reduced-value curvature, the new transfer law predicts the chosen observable, and the incidence experiment determines whether that observable effect is identifiable.

This positions the paper as the experimental outer layer of a four-paper program. Information-induced training geometry supplies explicit visible metrics and minimal SPD completion (Li, 2026a). P1 supplies finite-dimensional hidden-state reduction and reduced-value curvature (Li, 2026b). P2 supplies structured path-space geometry, Jacobi response, and its global Möbius realization (Li, 2026c). The present paper contributes the pathwise causal state, observable-readout transfer, experimental gauge, inference, and design layers. Its universal representation theorem does not require the regular companion theories.

Contributions.

1. We prove a universal causal–interventional representation theorem. Under the protocol's fixed innovation coupling, pathwise predictive equivalence gives the canonical behaviorally minimal optimizer state. Lower-finite incidence coordinates give unique pure effects, the complete design gauge, exact identifiability, sharp quotient stability, held-out falsification, and exact noiseless configuration complexity for every finite support.

2. Taking the P1 Schur–Möbius theorem as a structural input, we prove an observable readout interaction law. It transfers first and second hidden relaxation responses to arbitrary smooth scalar readouts, covers actual update coordinates, integrates exactly to their Boolean effects, and shows why the reduced-value sign law does not extend to general optimizer traces.

3. We develop the identifiable observation layer: projection certificates, Gaussian quotient minimax risk, exact confidence sets, residualized sector tests, misspecification decomposition, certified downstream actions, total-recovery allocation, and exact target-subspace A-optimal replication. The unrestricted Boolean recovery risk is exponential, quantifying the price of universal attribution.

4. We complete a controlled real-data factorial closure on a 65-dimensional strongly convex logistic hidden problem. Independent Boolean and curvature paths, continuous held-out intensities, eight data splits, information-optimal allocation, 100,000 Gaussian campaigns, 4500 real-minibatch observations, 20,000

bootstrap draws, held-out support rejection, and finite-response bounds validate the reduced-value structural input and the identification/inference layer. Separate neural audits instantiate full-SPD, diagonal, block, channel-scalar, layer-scalar, and trajectory-restricted response classes in nonconvex training.

Scope. Universality concerns pathwise causal representation under a fixed innovation coupling and incidence identification of expected responses. Smooth readout interaction requires the stated regularity, uniqueness, and positive vertical-Hessian assumptions. Exact Gaussian inference additionally assumes a finite-dimensional response and known positive-definite covariance. Trace explainability is not an optimizer-performance ranking.

2 Related Work and Series Position

Optimizer geometry and structured preconditioning. Riemannian optimization, natural gradient, and mirror descent formalize metric or dual-geometric update rules (Absil et al., 2008; Amari, 1998; Beck & Teboulle, 2003; Boumal, 2023). Numerical optimization develops variable metrics, quasi-Newton updates, line search, and trust regions (Nocedal & Wright, 2006; Combettes & Vũ, 2014; Parikh & Boyd, 2014). AdaGrad, Adam, K-FAC, Shampoo, Adafactor, SM3, and SOAP instantiate diagonal, block, Kronecker, factored, compressed, or moving-basis restrictions (Duchi et al., 2011; Kingma & Ba, 2015; Martens & Grosse, 2015; Gupta et al., 2018; Shazeer & Stern, 2018; Anil et al., 2019; Vyas et al., 2024). We treat each declared family as a response class or intervention sector.

Dynamic and learned optimizers. Feedback-system analysis, performance estimation, learning to optimize, and algorithm selection study optimizer behavior, synthesis, or benchmarking (Lessard et al., 2016; Drori & Teboulle, 2014; Taylor et al., 2017; Andrychowicz et al., 2016; Metz et al., 2022; Kerschke et al., 2019; Dahl et al., 2023). Our object is the counterfactual response law over declared configurations and the information available for identifying its effects.

Causal realization and system identification. Predictive equivalence follows the minimal-realization principle that two histories are merged exactly when no admissible future distinguishes them (Nerode, 1958). Predictive-state representations encode dynamical state directly through predictions of future observable tests (Singh et al., 2004). System identification reconstructs dynamics from controlled input–output observations (Ljung, 1999), while inverse optimization recovers latent objectives or parameters from decisions (Ahuja & Orlin, 2001). Our finite-horizon theorem uses the stronger pathwise equivalence induced by the protocol's fixed innovation coupling; this yields an exact right congruence, then connects the quotient to a finite-poset intervention gauge. It does not claim a distribution-only minimal stochastic realization.

Möbius effects and factorial experiments. Möbius inversion in incidence algebras is classical. Potential-outcome analyses of 2^K factorial designs define causal main and interaction effects and their randomization-based inference (Dasgupta et al., 2015). Shapley–Taylor indices and Integrated Hessians connect discrete feature interactions to derivative integrals under different axioms and baselines (Dhamdhere et al., 2020; Janizek et al., 2021). Randomization tests provide exact finite-sample calibration under invariant nulls (Hemerik & Goeman, 2018; Phipson & Smyth, 2010). Relative to classical factorial potential outcomes, we allow lower-finite posets and declared sparse supports, characterize the complete observation kernel and exact noiseless configuration complexity, and connect observable effects to a hidden optimizer readout law. We do not claim Möbius inversion or factorial causal contrasts themselves as new.

Parametric optimization sensitivity. Envelope and Schur-complement sensitivity formulas are standard in perturbation analysis and variable projection (Golub & Pereyra, 1973; Bonnans & Shapiro, 2000). Differentiable optimization layers, implicit differentiation, and bilevel learning compute related solution-map responses for optimization-defined modules (Amos & Kolter, 2017; Blondel et al., 2022; Franceschi et al., 2018). P1 specializes these ideas to hidden optimizer reduction, proves the negative-semidefinite affine-intervention Gram law, its factorial integral, and the noncommutation example restated here (Li, 2026b); P2 proves the path-space Möbius–Jacobi counterpart (Li, 2026c). Our new smooth result differentiates a general observable readout through the hidden solution map, exposing the additional terms that govern actual update interactions. The hidden factorization remains structural and is not identified from the discrete effect alone.

Experimental design. Classical optimal-design theory chooses measurements through information criteria and approximate design weights (Pukelsheim, 2006). Our incidence experiment specializes this perspective to optimizer intervention masks: rank gives exact effect identifiability, residualized columns give sector tests, and the saturated Boolean case admits a closed-form continuous A-optimal replication allocation. Integer rounding remains a separate design problem.

Mechanisms beyond positive geometry. Momentum, acceleration, clipping, signs, stochastic oracles, proximal maps, and schedules can produce behavior outside a positive parameter-space cometric applied to the current objective covector (Polyak, 1964; Nesterov, 1983; Su et al., 2016; Wibisono et al., 2016; Liu et al., 2023; Defazio et al., 2024). Muon-style rules apply matrix operators to momentum (Jordan, 2024; Essential AI et al., 2025; Crawshaw et al., 2025). Our protocol keeps such mechanisms as explicit sectors unless a different enlarged state geometry is declared.

Dependency within the series. The pathwise causal representation, Möbius identification, observable-readout transfer, and statistical results here are self-contained. The companion information-geometry paper supplies the explicit SPD metric-submetry realization (Li, 2026a); P1 supplies the finite-dimensional Schur and reduced-value interaction theorems (Li, 2026b); and P2 supplies path-space value functions, Jacobi response, and the global Möbius–Jacobi theorem (Li, 2026c). P3 imports those structural laws and contributes the experimental interface rather than republishing them as new results.

3 Three Equations for Causal Optimizer Interaction

3.1 Causal response

Let $(\mathcal{P}, \preceq)$ be lower-finite: every principal ideal is finite. For each configuration $a \in \mathcal{P}$, a finite-step causal optimizer experiment admits a history-state realization

$$x^a_{k+1} = \Phi^a_k(x^a_k, \mathcal{O}_f(x^a_k), \xi_k)\,, \qquad F(a) = \mathbb{E}\,\mathcal{R}(x^a_{0:N}) \in \mathbb{H}. \tag{1}$$

The protocol fixes initialization, data law, horizon, response map, upstream state, and coupling of exogenous randomness. The configuration changes only declared mechanisms. The response may contain update traces, logged states, terminal parameters, and task observables. Histories, inputs, exposed innovations, and responses are assumed standard Borel. The fixed innovation coupling is part of the protocol.

An *admissible pathwise realization* has a reachable state that is a deterministic function of the extended history, a response map depending only on that state, the declared input, and the current innovation, and a unifilar update from the same variables. Two reachable extended histories at the same time are *pathwise predictively equivalent* when every common future sequence of declared inputs and innovation values produces the same future response sequence. A realization is *behaviorally minimal* when it has no distinct reachable states with this equivalence. The quotient below is taken in this reachable pathwise/unifilar category. It is relative to the declared innovation representation and coupling. A distribution-only minimal state modulo null sets is a different measurable-kernel problem and is not claimed here.

The Hilbert-valued response in equation (1) and the scalar response used later have a precise interface. For every continuous linear functional $\varphi \in \mathbb{H}^*$ define $F_\varphi(a) = \varphi(F(a))$. The smooth reduction theorem applies to F_φ when this scalar observable has a regular reduced-value representation. A vector response can be treated componentwise only when each selected component has such a representation.

3.2 Pure interventional effects

Let $\mu_{\mathcal{P}}$ be the Möbius function. Define

$$\zeta_t = \sum_{s \preceq t} \mu_{\mathcal{P}}(s,t) F(s), \qquad F(a) = \sum_{t \preceq a} \zeta_t. \tag{2}$$

For $\mathcal{P} = 2^{[m]}$,

$$\zeta_T = \sum_{B \subseteq T} (-1)^{|T|-|B|} F(B), \qquad F(A) = \sum_{T \subseteq A} \zeta_T.$$

The baseline is $\zeta_\varnothing$, singleton effects are main effects, and larger sets are irreducible factorial interactions under the fixed protocol. This is a decomposition of counterfactual responses; it does not assert that the implementation internally adds vectors named ζ_T.

A finite model declares an effect support $\mathcal{K} \subseteq \mathcal{P}$ and sets $\zeta_t = 0$ outside $\mathcal{K}$. Main-effect, order-q, sparse hierarchical, and graded models are choices of $\mathcal{K}$. They are falsifiable because fitted effects predict held-out configurations.

3.3 Observation and information

For queried configurations $\mathcal{D} = (a_1, \ldots, a_n)$, define

$$X_{\mathcal{D},\mathcal{K}}(i,t) = \mathbf{1}\{t \preceq a_i\}.$$

When $\mathbb{H} \cong \mathbb{R}^d$, the observation equation is

$$Y = \mathcal{A}_\mathcal{D}\zeta + \varepsilon, \qquad \mathcal{A}_\mathcal{D} = X_{\mathcal{D},\mathcal{K}} \otimes I_d, \qquad \mathcal{I} = \mathcal{A}_\mathcal{D}^* \Sigma^{-1} \mathcal{A}_\mathcal{D}. \tag{3}$$

The kernel of $\mathcal{A}_\mathcal{D}$ is the observational gauge. Its positive singular spectrum controls stable recovery; the information operator controls finite-sample inference and design.

3.4 Geometric response classes

For a visible covector α_k, let $Q_k = \eta_k P_k$ be the effective positive map and $u_k = -Q_k\alpha_k$. Absorbing the step scale into Q_k prevents temporal audits from hiding arbitrary variation in η_k. Fix a finite-dimensional normed parameterization $\mathbb{Q}$ of the effective maps along the audited trace, using a declared trivialization when tangent spaces vary, and define

$$\mathrm{Var}(Q_{0:N-1}) = \sum_{k=0}^{N-2} \|Q_{k+1} - Q_k\|_\mathbb{Q}.$$

For a family $\mathcal{F}$ and path budget B, define

$$\mathcal{M}_{\mathcal{F},B} = \left\{(-Q_k\alpha_k)_{k=0}^{N-1} : Q_k \in \mathcal{P}_\mathcal{F}(\theta_k),\ \mathrm{Var}(Q_{0:N-1}) \le B\right\} \subseteq \mathbb{H}. \tag{4}$$

Distance to this set tests membership in a declared response class. Interventions identify counterfactual coordinates. These operations share a response space and answer different questions.

4 Universal Causal–Interventional Representation

Theorem 1 (Universal causal–interventional optimizer representation)**.** *Let $(\mathcal{P}, \preceq)$ be lower-finite, let $F : \mathcal{P} \to \mathbb{H}$ be induced by equation* (1)*, and let $\mathcal{K} \subseteq \mathcal{P}$ and $\mathcal{D}$ be finite effect-support and design families.*

(i) Minimal causal state. *Every innovation-driven causal optimizer under the fixed protocol has a history-state realization. Pathwise predictive-equivalence classes form a sufficient pathwise/unifilar realization. Every other admissible sufficient reachable pathwise realization maps onto this quotient, and every behaviorally minimal reachable realization is isomorphic to it as a reachable transition–response system.*

(ii) Incidence completeness. *Equation* (2) *is the unique incidence-algebra expansion of F.*

(iii) Complete experimental gauge. *Two supported effect collections are observationally equivalent exactly when*

$$\zeta - \zeta' \in \ker(X_{\mathcal{D},\mathcal{K}} \otimes I_\mathbb{H}).$$

If $\dim \mathbb{H} = d$ and $r_\mathcal{D} = \operatorname{rank} X_{\mathcal{D},\mathcal{K}}$, every compatible fiber has dimension $d(|\mathcal{K}|-r_\mathcal{D})$. Exact identification holds if and only if $r_\mathcal{D} = |\mathcal{K}|$.

(iv) Sharp quotient stability. *If $X = X_{\mathcal{D},\mathcal{K}}$ has positive rank and $y = (X \otimes I_d)\zeta + e$, the minimum-norm inverse estimates the canonical representative $\zeta_{\mathrm{can}} \in \ker(X \otimes I_d)^\perp$ with*

$$\left\|\widehat{\zeta} - \zeta_{\mathrm{can}}\right\| \le \frac{\left\|\mathrm{Proj}_{\mathrm{range}(X\otimes I_d)}\, e\right\|}{\sigma^+_{\min}(X)}.$$

The constant is sharp.

(v) Exact intervention complexity. *Suppose $\mathbb{H} \neq \{0\}$ and the supported class contains every collection in $\mathbb{H}^{\mathcal{K}}$. In a noiseless expected-response oracle model, every fixed, deterministic adaptive, or randomized adaptive scheme that exactly recovers every supported law with probability one needs at least $|\mathcal{K}|$ distinct configuration queries in the worst case. Querying $\mathcal{D} = \mathcal{K}$ attains the bound:*

$$N_{\mathrm{exact}}(\mathcal{K}) = |\mathcal{K}|.$$

(vi) Prediction and falsification. *If $X_{\mathcal{D},\mathcal{K}}$ has full column rank, the recovered effects predict*

$$F_{\mathcal{K}}(b) = \sum_{t\in\mathcal{K}:t\preceq b} \zeta_t \qquad \textit{for every } b \in \mathcal{P}.$$

A held-out response inconsistent with this equation rejects the support model under the fixed protocol.

The proof is in section A. The randomized lower bound uses the finite set of incidence row patterns induced by $\mathcal{K}$; it does not require the ambient lower-finite poset itself to be finite. The count N_{exact} is configuration complexity, not the number of noisy optimizer runs. Replication under observation noise belongs to Theorem 4.

Boolean and graded complexity. For $\mathcal{P} = 2^{[m]}$, unrestricted attribution costs 2^m configurations. Under the order-q support

$$\mathcal{K}_q = \{T \subseteq [m] : |T| \le q\}, \qquad N_q = \sum_{j=0}^{q} \binom{m}{j}.$$

Product-of-chain interventions yield graded mixed differences, with exact cost equal to the retained multi-index count.

Passive impossibility. If a finite poset has a greatest configuration $\top$, the one-row passive design $\mathcal{D} = \{\top\}$ has rank one. It leaves $d(|\mathcal{K}| - 1)$ unidentifiable coordinates. A passive geometric residual can exclude a declared class, but universal causal attribution requires interventions or an equivalent injective observation.

No universal dissipative representation. The theorem includes causal rules with cycles. For example, $x_{k+1} = x_k + 1 \pmod 3$ is causal but cannot satisfy a strict scalar Lyapunov law, since it would imply $V(0) > V(1) > V(2) > V(0)$. Smooth gradient or Onsager dynamics are regular subclasses of the causal response theory.

5 Hidden Relaxation Generates Observable Interaction

The universal theorem identifies effects without assigning them a smooth mechanism. The companion variational-reduction paper P1 already proves the hidden-response, Schur-curvature, affine-intervention, factorial-integral, and noncommutation results used below (Li, 2026b). We restate that structural input in the present notation, then derive a new readout-transfer law that applies to observed optimizer updates rather than only to the reduced optimal value.

Let $\mathcal{Y}$ and $\mathcal{S}$ be finite-dimensional smooth manifolds, let $y \in \mathcal{Y}$ be visible, $\sigma \in \mathcal{S}$ hidden, and let $u \in U \subset \mathbb{R}^m$ be a continuous intervention amplitude, where U is open and contains $[0,1]^m$. Derivatives below are coordinate matrices in a chosen local product chart; on manifolds they may instead be read as covariant derivatives for chosen connections. The u block is unambiguous in the declared Euclidean intervention coordinates. Because

$E_\sigma = 0$ at the minimizing section, the Schur expression is invariant under smooth reparameterizations of the hidden coordinate. Consider a scalar response generated by

$$\bar{E}(y,u) = \inf_{\sigma \in \mathcal{S}} E(y,\sigma,u). \tag{5}$$

Assumption 1 (Regular hidden relaxation)**.** On $\mathcal{Y} \times U$, the energy is C^{r+1} for some $r \geq 2$ and is locally uniformly inf-compact in σ: over every compact parameter set, each finite fiber sublevel union is relatively compact. Every $(y,u) \in \mathcal{Y} \times U$ has a unique interior minimizer $\sigma_\star(y,u)$, and

$$H(y,u) = D^2_{\sigma\sigma} E(y, \sigma_\star(y,u), u)$$

is positive definite.

Theorem 2 (Reduction-induced optimizer interaction; P1, restated)**.** *Under Assumption 1, the minimizer section is C^r and*

$$D_{(y,u)}\sigma_\star = -H^{-1} E_{\sigma(y,u)}, \tag{6}$$

$$D^2_{(y,u)(y,u)}\bar{E} = E_{(y,u)(y,u)} - E_{(y,u)\sigma} H^{-1} E_{\sigma(y,u)}. \tag{7}$$

If interventions enter affinely before reduction,

$$E(y,\sigma,u) = E_0(y,\sigma) + \sum_{i=1}^{m} u_i \Delta_i(y,\sigma), \tag{8}$$

and $G : \mathbb{R}^m \to T^*_\sigma \mathcal{S}$ is $Ga = \sum_i a_i D_\sigma \Delta_i$, then

$$\boxed{D^2_{uu}\bar{E} = -G^* H^{-1} G \preceq 0.} \tag{9}$$

For Boolean vertex responses $F(A) = \bar{E}(y, \mathbf{1}_A)$ and every $T \subseteq [m]$ with $|T| \leq r$,

$$\zeta_T = \int_{[0,1]^T} \partial_T^{|T|} \bar{E}\left(y, \sum_{i\in T} t_i e_i\right) \prod_{i \in T} \mathrm{d}t_i. \tag{10}$$

In particular,

$$\boxed{\zeta_{\{i,j\}} = -\int_0^1 \int_0^1 \left\langle D_\sigma \Delta_i, H^{-1} D_\sigma \Delta_j \right\rangle_{(se_i + te_j, \sigma_\star)} \,\mathrm{d}s\,\mathrm{d}t.} \tag{11}$$

Equation (9) is a negative-semidefinite Gram law. A pointwise mixed response vanishes exactly when the corresponding vertical perturbations are orthogonal in the inverse-Hessian metric. A zero integrated Möbius effect may also result from cancellation, so the converse need not hold for (11).

The bridge is a forward structural law. A factorial experiment identifies its left-hand side, but the factorization into H and G is generally nonunique. A nonzero pair effect therefore does not by itself identify hidden geometry, inverse stiffness, or a particular optimizer state. Testing the right-hand side requires an independently specified and measured hidden model.

The reduced value is only one possible experimental response. Actual optimizer traces usually observe a readout of the relaxed state. The next result gives the missing interface.

Theorem 3 (Observable readout interaction transfer)**.** *Under Assumption 1, fix y and let $R : U \times \mathcal{S} \to \mathbb{R}$ be C^2. Define the observed scalar response*

$$F_R(u) = R(u, \sigma_\star(u)).$$

In the chosen hidden chart, put

$$v_i = \partial_{u_i}\sigma_\star = -H^{-1} E_{\sigma u_i}$$

and

$$w_{ij} = -H^{-1}\big(E_{\sigma\sigma\sigma}[v_i, v_j] + E_{\sigma\sigma u_i} v_j + E_{\sigma\sigma u_j} v_i + E_{\sigma u_i u_j}\big)\,. \tag{12}$$

Then $w_{ij} = \partial^2_{u_i u_j}\sigma_\star$ *and*

$$\boxed{\partial^2_{u_i u_j} F_R = R_{u_i u_j} + R_{u_i \sigma} v_j + R_{\sigma u_j} v_i + R_{\sigma\sigma}[v_i, v_j] + R_\sigma w_{ij}.} \tag{13}$$

Consequently the Boolean pair effect of the actual readout is

$$\zeta^R_{\{i,j\}} = \int_0^1 \int_0^1 \partial^2_{u_i u_j} F_R(s e_i + t e_j)\, \mathrm{d}s\, \mathrm{d}t. \tag{14}$$

For affine interventions, $E_{\sigma u_i u_j} = 0$ *and* $E_{\sigma\sigma u_i} = D^2_{\sigma\sigma}\Delta_i$*. Unlike the reduced-value law, a general readout need not have negative-semidefinite interaction curvature. Hilbert-valued updates are covered after composition with any declared scalar functional* $\varphi \in \mathbb{H}^*$*.*

Proposition 1 (Reduction and Möbius transformation do not commute)**.** *There is no general identity between taking a hidden infimum before a Möbius transform and taking an infimum of pointwise Möbius differences.*

Indeed, for $E_0(\sigma) = \sigma^2$ and $E_1(\sigma) = (\sigma - 1)^2$, both reduced values are zero, so the reduced one-sector effect is zero. The pointwise difference is $1 - 2\sigma$, whose infimum is $-\infty$. Fiber-constant intervention increments are a sufficient commuting case: they leave $\sigma_\star$ unchanged and create no higher-order effects.

Exact analytical example. Let

$$E(\sigma, u) = \frac{h}{2}\sigma^2 + u_1 a\sigma + u_2 b\sigma, \qquad h > 0.$$

Then $\sigma_\star = -(au_1 + bu_2)/h$ and

$$\bar{E}(u) = -\frac{(au_1 + bu_2)^2}{2h}, \qquad \zeta_{\{1,2\}} = -\frac{ab}{h}.$$

The interaction is large when hidden relaxation is compliant (h small), and its sign records alignment of the two vertical perturbations.

Coupled adaptive diagonal-preconditioner example. Let $g = (g_1, g_2)$ be the visible gradient and let $\sigma \in \mathbb{R}^2$ be a hidden log-second-moment state selecting

$$P(\sigma) = \operatorname{diag}(e^{-\sigma_1/2}, e^{-\sigma_2/2}), \qquad q(\sigma, g) = -P(\sigma) g.$$

For $h > |\rho|$, a statistic $b(g) \in \mathbb{R}^2$, and two declared protocol increments $\Delta_1(\sigma) = \sigma_1$, $\Delta_2(\sigma) = \sigma_2$, consider the regularized state-selection energy

$$E(g, \sigma, u) = \frac{1}{2}\sigma^\top \underbrace{\begin{pmatrix} h & \rho \\ \rho & h \end{pmatrix}}_{H} \sigma - b(g)^\top \sigma + u_1 \Delta_1(\sigma) + u_2 \Delta_2(\sigma). \tag{15}$$

Here $G = [e_1\ e_2] = I_2$, $\sigma_\star = H^{-1}(b(g) - u)$, and the scalar observable is the optimized calibration value $F(A) = \bar{E}(g, \mathbf{1}_A)$. The Boolean pair effect is

$$\zeta_{\{1,2\}} = -e_1^\top H^{-1} e_2 = \frac{\rho}{h^2 - \rho^2}. \tag{16}$$

Thus interventions acting on different diagonal channels interact whenever the hidden adaptation penalty couples those channels. More directly, the first coordinate of the actual optimizer update is a readout

$$q_1(u) = -g_1 e^{-\sigma_{\star,1}(u)/2} = q_1(0) \exp\!\left(\frac{h}{2(h^2 - \rho^2)} u_1 - \frac{\rho}{2(h^2 - \rho^2)} u_2\right).$$

Its identifiable Boolean pair effect is

$$\zeta^{q_1}_{\{1,2\}} = q_1(0)\left(e^{\frac{h}{2(h^2-\rho^2)}} - 1\right)\left(e^{-\frac{\rho}{2(h^2-\rho^2)}} - 1\right), \tag{17}$$

which is nonzero when $g_1\rho \neq 0$. The calibration-value contrast in equation (16) follows the P1 Gram law; the observed-update contrast follows the new readout law and has no universal sign. This example specifies the optimizer state, selected positive geometry, energy, intervention increments, vertical Hessian, perturbation map, reduced response, actual update response, and both predicted factorial contrasts. Its full calculation is in section B.

Structured regular subclasses. For information-induced SPD completion, H^{-1} is the inverse vertical curvature of the completion energy supplied by P4 (Li, 2026a). P1 supplies the finite-dimensional reduction and reduced-value interaction laws restated above (Li, 2026b). P2 supplies the path-space Jacobi–Schur and Möbius–Jacobi versions (Li, 2026c). The new readout theorem transports either hidden response to experimentally chosen update or trace observables. These realizations require their own regularity assumptions and do not restrict Theorem 1.

6 Observation, Inference, and Experimental Design

Let $\mathbb{Y} \cong \mathbb{R}^n$ have covariance $V \succ 0$ and norm $\|y\|^2_{V^{-1}} = y^\top V^{-1} y$. Let $\mathcal{M} \subseteq \mathbb{Y}$ be a nonempty closed convex response class, let $\mathbb{Z}$ be a finite-dimensional real inner-product space, and let the linear map $A : \mathbb{Z} \to \mathbb{Y}$ generate responses from coordinates $z \in \mathbb{Z}$. Write

$$\mathcal{I} = A^* V^{-1} A, \qquad K = \ker A, \qquad r = \operatorname{rank} A.$$

Theorem 4 (Identifiable observation–inference–decision calculus)**.** *Under this setup:*

(i) Every y has a unique shadow $\widehat{m} = \arg\min_{m\in\mathcal{M}} \frac{1}{2}\|y-m\|^2_{V^{-1}}$. Define $\delta_{\mathcal{M}}(y) = \|y-\widehat{m}\|_{V^{-1}}$. With the extended-real support function $\sigma_{\mathcal{M}}(q) = \sup_{m\in\mathcal{M}} q^\top m \in \mathbb{R}\cup\{+\infty\}$,

$$\frac{1}{2}\delta_{\mathcal{M}}(y)^2 = \max_q \left\{ q^\top y - \sigma_{\mathcal{M}}(q) - \frac{1}{2} q^\top V q \right\}, \tag{18}$$

and $q^\star = V^{-1}(y-\widehat{m})$. Distance is one-Lipschitz and decreases under nesting of response classes.

(ii) In $Y = Az + \varepsilon$, $\varepsilon \sim N(0,V)$, the identifiable parameter is the quotient by K. On $K^\perp$, generalized least squares

$$\widehat{z} = \mathcal{I}^\dagger A^* V^{-1} Y$$

is exactly minimax:

$$\inf_{\widetilde{z}} \sup_{z\in K^\perp} \mathbb{E}_z \|\widetilde{z} - z\|^2 = \operatorname{tr}(\mathcal{I}^\dagger), \tag{19}$$

and $(\widehat{z}-z)^*\mathcal{I}(\widehat{z}-z) \sim \chi^2_r$.

(iii) Partition $A = [A_{-G}\ A_G]$, put $\widetilde{A} = V^{-1/2}A$, let Π_{-G} project onto $\operatorname{range}(\widetilde{A}_{-G})$, and define $B_G = (I-\Pi_{-G})\widetilde{A}_G$. The statistic

$$\left\| \operatorname{Proj}_{\operatorname{range}(B_G)} V^{-1/2} Y \right\|^2$$

is central $\chi^2_{\operatorname{rank} B_G}$ under $B_G z_G = 0$ and noncentral with parameter $\|B_G z_G\|^2$ otherwise. If the true mean is $Az + b$ with the canonical representative $z \in K^\perp$, let Π_A project onto $\operatorname{range}(V^{-1/2}A)$ and define

$$b_\parallel = \mathcal{I}^\dagger A^* V^{-1} b, \qquad b_\perp = (I-\Pi_A)V^{-1/2} b.$$

Omitted structure decomposes exactly into estimation bias and orthogonal lack of fit:

$$\mathbb{E}\|\widehat{z}-z\|^2 = \left\|b_\parallel\right\|^2 + \operatorname{tr}(\mathcal{I}^\dagger), \qquad \left\|(I-\Pi_A)V^{-1/2}Y\right\|^2 \sim \chi^2_{n-r}(\|b_\perp\|^2). \tag{20}$$

(iv) Every linear downstream map obeys

$$\|L(y-\widehat{m})\| \le \left\|LV^{1/2}\right\|_{\mathrm{op}} \delta_{\mathcal{M}}(y).$$

For a declared finite action class, or more generally an action class on which the displayed minima are attained, let $\Delta(\pi;z) = d_\pi + \ell_\pi^\top z$ *with* $\ell_\pi \in K^\perp$ *and define*

$$\rho_\pi(\delta) = \sqrt{\chi^2_{r,1-\delta}}\sqrt{\ell_\pi^\top \mathcal{I}^\dagger \ell_\pi}, \qquad \widehat{\pi} \in \arg\min_\pi \{\Delta(\pi;\widehat{z}) + \rho_\pi(\delta)\}.$$

On one event of probability $1-\delta$,

$$\Delta(\widehat{\pi};z) \le \Delta(\pi^\star;z) + 2\rho_{\pi^\star}(\delta), \qquad \pi^\star \in \arg\min_\pi \Delta(\pi;z). \tag{21}$$

(v) For a saturated incidence design $\mathcal{D} = \mathcal{K}$, *let* $M = X^{-1}_{\mathcal{K},\mathcal{K}}$. *With independent response covariance* Σ *and continuous design weights* $n_a > 0$, $\sum_a n_a = N$, *the exact relaxed total recovery risk is*

$$R(n) = \mathrm{tr}(\Sigma)\sum_a \frac{c_a}{n_a}, \qquad c_a = \|M_{:,a}\|_2^2, \tag{22}$$

with unique optimum

$$n_a^\star = N\frac{\sqrt{c_a}}{\sum_b \sqrt{c_b}}, \qquad R^\star = \frac{\mathrm{tr}(\Sigma)}{N}\left(\sum_a \sqrt{c_a}\right)^2. \tag{23}$$

For the unrestricted Boolean lattice,

$$c_A = 2^{m-|A|}, \quad n_A^\star = N\frac{2^{(m-|A|)/2}}{(1+\sqrt{2})^m}, \quad R^\star = \frac{\mathrm{tr}(\Sigma)}{N}(3+2\sqrt{2})^m. \tag{24}$$

These formulas solve the continuous allocation problem exactly. Integer replication counts require a feasible rounding rule and its associated excess-risk analysis.

Corollary 1 (Exact target-optimal replication)**.** *In the saturated scalar incidence design, let* $M = X^{-1}_{\mathcal{K},\mathcal{K}}$ *and let* L *select or linearly combine target effects. Suppose the independent configuration-level observation variance is* σ_a^2/n_a *and*

$$c_a = \|LM_{:,a}\|_2^2 > 0.$$

Then the exact target risk and its continuous optimum are

$$R_L(n) = \mathrm{tr}\left[LM\,\mathrm{diag}\left(\frac{\sigma_a^2}{n_a}\right)M^\top L^\top\right] = \sum_a \frac{c_a\sigma_a^2}{n_a}, \tag{25}$$

$$n_a^\star = N\frac{\sigma_a\sqrt{c_a}}{\sum_b \sigma_b\sqrt{c_b}}, \qquad R_L^\star = \frac{1}{N}\left(\sum_a \sigma_a\sqrt{c_a}\right)^2. \tag{26}$$

The integer design used in section 8 rounds these weights while preserving the total budget and a declared minimum replication.

The proof is in section C. The theorem turns the deterministic gauge of Theorem 1 into finite-sample uncertainty. Rank determines what is identifiable; the positive spectrum determines how accurately; the target decision determines which directions matter.

Exact non-Gaussian calibration. If a finite group G leaves the null response law invariant, then for any residual statistic T,

$$p_G(Y) = \frac{1}{|G|}\sum_{g\in G} \mathbf{1}\{T(gY) \ge T(Y)\} \tag{27}$$

is super-uniform under the null (Hemerik & Goeman, 2018; Phipson & Smyth, 2010). This calibrates arbitrary response-class residuals when the group action is scientifically justified.

Design for interaction curvature. For the mask experiment $A = X_{\mathcal{D},\mathcal{K}} \otimes I_d$, design can maximize rank, the smallest positive information eigenvalue, or a target contrast precision. To identify the left-hand side of equation (11), the support must include the relevant pair effect and the queried masks must make its residualized column nonzero. An order-two design with that rank property is the minimal direct identification extension; separate held-out higher masks test the order-two support assumption. Validation of the hidden right-hand side additionally needs measurements or a separately fitted structural model for H and G.

7 Structured Geometry as a Response Class

Geometry is one declared sector inside the causal calculus. This section records the response classes used by the experiments and clarifies their representational boundaries.

7.1 Pointwise expressivity

Proposition 2 (Full-SPD geometricization). *For nonzero $g \in \mathbb{R}^d$ and $u \in \mathbb{R}^d$, there exists $P \in \mathbb{S}^d_{++}$ with $u = -Pg$ if and only if $g^\top u < 0$.*

Thus full positive geometry expresses exactly strict descent directions relative to the declared visible covector. At an exact critical point, every pure positive-geometric update is zero.

Corollary 2 (Critical-point boundary). *If $g = 0$ and $u \neq 0$, the motion requires another declared sector, an enlarged state-space geometry, or a non-gradient mechanism.*

For the positive diagonal family $P = \mathrm{diag}(p_1, \ldots, p_d)$, $p_i > 0$:

Proposition 3 (Diagonal expressivity and residual). *Exact expression holds if and only if $g_i = 0 \Rightarrow u_i = 0$ and $g_i \neq 0 \Rightarrow u_i g_i < 0$ for every i. Moreover,*

$$\rho_{\mathrm{diag}}(u; g)^2 = \sum_{i:g_i=0} u_i^2 + \sum_{i:g_i\neq 0,\, u_i g_i > 0} u_i^2.$$

If $g_i \neq 0$ and $u_i = 0$, the coordinate infimum is zero but is not attained in the open positive cone.

For a coordinate partition $\{B_1, \ldots, B_s\}$:

Proposition 4 (Block expressivity). *A block-diagonal $P \succ 0$ satisfies $u = -Pg$ if and only if each block obeys $g_{B_j} = 0 \Rightarrow u_{B_j} = 0$ and $g_{B_j} \neq 0 \Rightarrow g_{B_j}^\top u_{B_j} < 0$.*

The proofs and the closed-form bounded-diagonal projection are in section D. Diagonal and block statements depend on parameterization; intrinsic claims require an intrinsic reference geometry and family.

7.2 Trajectory residual complexity

For a trace $\mathcal{T}_N = \{(\theta_k, \alpha_k, u_k)\}_{k=0}^{N-1}$ with $\|u\| > 0$, stack the updates in the product response space and define

$$\mathrm{TRC}_{\mathcal{F},B}(\mathcal{T}_N) = \inf_{m \in \mathcal{M}_{\mathcal{F},B}} \|u - m\|, \qquad \mathrm{TGER}_{\mathcal{F},B} = 1 - \frac{\mathrm{TRC}_{\mathcal{F},B}}{\|u\|}. \tag{28}$$

The raw residual permits a fresh map at each step and therefore audits a high-capacity response class. A coherent budget restricts temporal variation; $B = 0$ fits one shared map in the declared parameterization. Since the model class need not contain the zero response, $\mathrm{TGER} \leq 1$ may be negative; zero means that the best class member is no closer than the zero response under the chosen norm.

Closedness and convexity require structure. A sufficient condition for the audited finite traces is that every parameter fiber is nonempty compact convex in the common finite-dimensional space $\mathbb{Q}$, at least one parameter path satisfies all fiber constraints and the variation budget, $Q \mapsto -Q\alpha_k$ is linear, and Var is the convex continuous variation defined in section 3. The feasible parameter path set is then nonempty compact convex and its response image is nonempty compact convex. This covers the bounded-diagonal, bounded block, channel-scalar, and layer-scalar audits, whose shared constant maps provide budget-feasible paths. Open positive cones and arbitrary nonlinear SPD path families do not inherit this conclusion. Under the sufficient condition, Theorem 4 gives a unique shadow, a dual residual certificate, nesting monotonicity, and perturbation stability.

For the accumulation map $L_t(r_0, \ldots, r_{N-1}) = \sum_{k<t} r_k$,

$$\left\|\theta_t - \bar{\theta}_t\right\| \leq \|L_t\|_{\mathrm{op}} \, \mathrm{TRC}_{\mathcal{F},B} \, .$$

Thus a response-space residual transfers to trajectory shadowing. It does not identify which implementation module caused the residual.

7.3 Relation to the companion geometry papers

P4 studies when a visible SPD action has a unique minimal full completion and supplies exact AIRM quotient geometry (Li, 2026a). P1 studies smooth hidden-variable reduction and metric evolution (Li, 2026b). P2 restricts geometry families and path budgets (Li, 2026c). Here their outputs become response classes and intervention coordinates. The causal and statistical theorems remain valid for nongeometric sectors as well.

8 Factorial Closure and Response-Class Audits

The evidence has two deliberately different layers. A controlled real-data experiment closes the complete chain from hidden strong convexity through Schur curvature, Möbius effects, experimental design, inference, falsification, and finite-response tracking. Neural-network trace audits then test whether the declared response classes remain informative in nonconvex training. Only the first layer is a direct validation of the reduced-value interaction law.

8.1 A real-data hidden optimization model

We use the handwritten digits 3 and 8 from the scikit-learn digits data. A stratified split gives 249 training and 108 test examples. After standardizing the 64 pixels and adding an intercept, the hidden state is $w \in \mathbb{R}^{65}$. The base energy is ridge logistic regression,

$$E_0(w) = \frac{1}{N}\sum_{n=1}^{N} \log(1 + e^{-y_n x_n^\top w}) + \frac{\lambda}{2}\|w\|^2, \qquad \lambda = 0.15.$$

Three continuous interventions $u \in [0,1]^3$ emphasize the digit-3 examples, the digit-8 examples, and a central-2×2-occlusion robustness loss. If Δ_i is the corresponding mean logistic loss, the hidden energy and declared response are

$$E(w,u) = E_0(w) + 0.75\sum_{i=1}^{3} u_i \Delta_i(w), \qquad F(u) = \min_w E(w,u). \tag{29}$$

Every added loss is convex, so $E_{ww} \succeq 0.15I$ on the whole intervention cube and every configuration has a unique global minimizer. The largest terminal gradient infinity norm across the eight Boolean solves is 3.09×10^{-13}.

8.2 Curvature, Boolean effects, and continuous prediction

Boolean effects are computed from eight independently optimized corner responses. Separately, order-24 Gauss–Legendre quadrature evaluates $-g_i^\top H^{-1} g_j$ after resolving the hidden optimum at every quadrature node. Neither path uses the output of the other.

Across the three pairs, the maximum Boolean identity error is 4.21×10^{-11}. We also reserve three non-Boolean intensity points per pair, $(0.2, 0.7)$, $(0.45, 0.35)$, and $(0.8, 0.55)$. Direct joint optimization and edge-response-plus-curvature reconstruction agree at all nine points with maximum error 8.88×10^{-13}. Repeating all

Table 1: Independent Boolean and curvature-integral evaluations for the three mechanism pairs in equation (29). Pair 1–2 emphasizes the two classes; pairs 1–3 and 2–3 combine class information with occlusion robustness.

Pair	Boolean effect	Curvature integral	Abs. error
1-2	+0.000625	+0.000625	8.36e-15
1-3	-0.007276	-0.007276	4.21e-11
2-3	-0.014264	-0.014264	3.16e-11

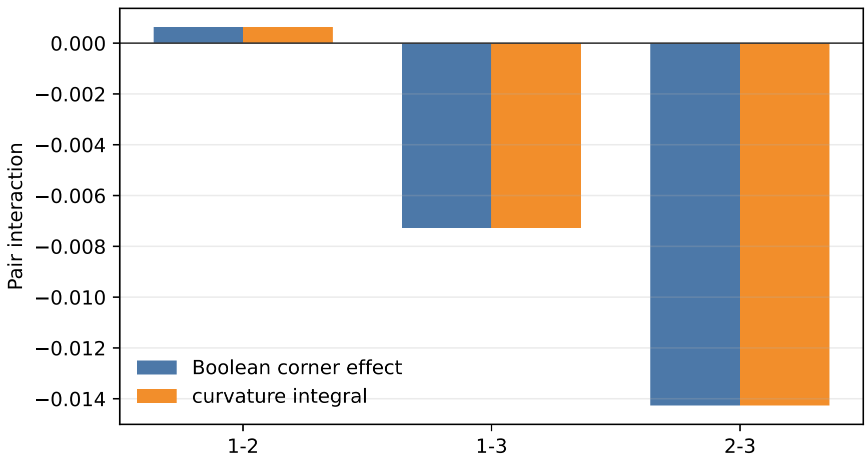

Figure 1: Boolean pair effects and independently integrated Schur curvature. The largest absolute discrepancy is 4.21×10^{-11}.

Boolean solves and three curvature integrals on eight independent stratified data splits gives 24 additional identities with maximum error 2.81×10^{-8}. The small class–class effect changes sign across splits, whereas both class–occlusion effects remain negative.

8.3 Information design and Gaussian confirmation

We declare the order-two support

$$\mathcal{K}_2 = \{\varnothing, 1, 2, 3, 12, 13, 23\}$$

and query its seven masks. Their incidence matrix is invertible. Mask 123 is held out to test the omitted third-order effect, whose exact value is

$$\zeta_{123} = 0.002366036.$$

By Corollary 1, the target risk for the three pair effects has allocation weights $\sigma_a \|LM_{:,a}\|$. For the declared independent Gaussian channel $Y_{a,r} = F(a) + \varepsilon_{a,r}$ with $\sigma_a = 0.004(1 + 0.2|a|)$, the pilot budget $(420, 80)$ for main and held-out replications has only 0.240 exact power. Scaling both budgets by the first integer that reaches the predeclared 0.95 target gives 3780 main and 720 held-out replications. Pair-A-optimal integer allocation is

$$(594, 582, 582, 582, 480, 480, 480).$$

In 100,000 independent confirmatory campaigns, exact theoretical power is 0.96309 and empirical rejection is 0.96260. Held-out null rejection is 0.04958, and the joint 95% pair ellipsoid covers with probability 0.94973. Pair-A-optimal theoretical and empirical risks are respectively 5.1433×10^{-7} and 5.1297×10^{-7}, both below the uniform-design values 5.1911×10^{-7} and 5.2050×10^{-7}. The improvement is only 0.92% under this mild heteroscedasticity; the point is agreement of the information calculation and the Monte Carlo experiment, not a large practical gain.

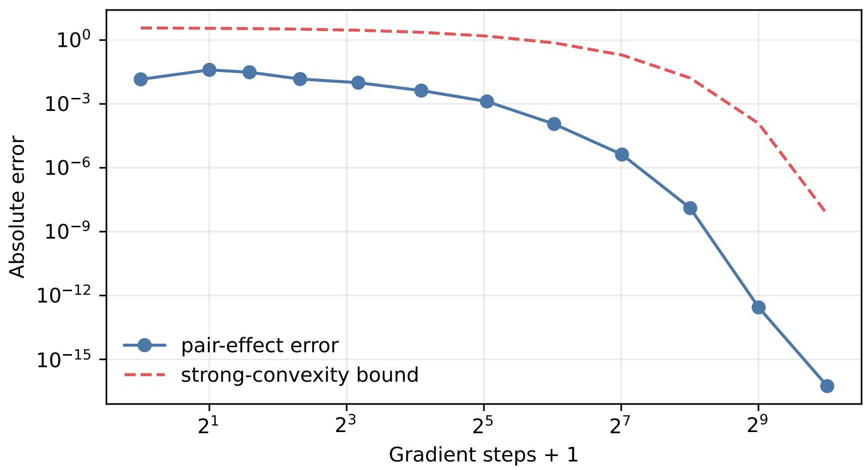

Figure 2: Convergence of finite-horizon pair effects to the quasistatic effects. The maximum error decreases from 1.43×10^{-2} at initialization to 5.55×10^{-17} after 1024 steps and stays below the theoretical bound.

8.4 A data-driven minibatch channel

The Gaussian channel tests the exact finite-sample theorem under its stated model. A second channel uses only real minibatch losses. At each optimized configuration, independently sampled minibatches provide an unbiased estimate of every active loss component in equation (29). A pilot of 200 observations per mask estimates heteroscedastic standard deviations and sets, before confirmation, the allocation

$$(624, 537, 614, 625, 386, 458, 536),$$

again with 3780 main and 720 held-out observations. All 4500 confirmatory raw responses are retained. Stratified nonparametric bootstrap with 20,000 draws gives

$$\begin{aligned} \zeta_{12}: &\quad -0.000251, \quad [-0.001094,\ 0.000601], \\ \zeta_{13}: &\quad -0.008050, \quad [-0.008900, -0.007226], \\ \zeta_{23}: &\quad -0.014293, \quad [-0.015164, -0.013418]. \end{aligned}$$

The two stable class–occlusion intervals contain their fixed-split truths and exclude zero. The small class–class interval includes zero and misses its fixed-split truth 0.000625 by 2.35×10^{-5}; we retain this miss rather than claiming simultaneous coverage from three marginal 95% intervals. The independent held-out triple estimate is 0.003114 with interval $[0.001839, 0.004399]$, rejecting the order-two support under real, heteroscedastic, non-Gaussian sampling noise. Bootstrap pair risk 5.7044×10^{-7} agrees with the information estimate 5.7297×10^{-7} and is below the uniform estimate 5.8620×10^{-7}.

8.5 Finite-response tracking

The Schur law is quasistatic, whereas an optimizer reaches $w_\star(u)$ in finite time. We run fixed-step gradient descent at the analytic smoothness bound. Strong convexity supplies a geometric objective-gap bound, and the sum of four corner gaps bounds every pair-effect error. At every audited horizon the actual error is below the exact gap bound, which is in turn below the strong-convexity bound.

8.6 Neural response-class audits

The controlled experiment validates the theorem under global strong convexity. We retain the earlier neural audits for a different purpose: testing whether declared geometric response classes distinguish optimizer traces in nonconvex training. A synthetic sanity check first separates fixed diagonal, time-varying diagonal, momentum, and operator-mixing mechanisms.

Table 2: Synthetic response-class sanity check. Raw TGER permits a step-specific diagonal map; coherent TGER fits one map over the trace.

Mechanism	Mean GER	Raw TGER	Coh. TGER	Gap
constant diagonal	1.000	1.000	1.000	0.000
varying diagonal	1.000	1.000	0.520	0.480
momentum memory	0.484	0.455	0.227	0.227
decoupled decay	0.925	0.924	0.761	0.163
operator mixing	0.785	0.785	0.494	0.291

Table 3: Trace-audit seed means. BDiag uses bounded-diagonal raw TGER under the raw-gradient convention; +WD changes the visible covector; Layer uses the layer-scalar family. Sample standard deviations are retained in the supplied CSV files.

Dataset	Opt.	Acc. (%)	BDiag	+WD covec.	Layer
MNIST	AdamW	97.6	0.274	0.256	0.086
	Sophia	97.5	0.187	0.278	0.029
	Muon-style	97.6	0.093	0.228	0.016
	GNG-Muon	97.9	0.101	0.232	0.015
Fashion-MNIST	AdamW	88.6	0.400	0.364	0.108
	Sophia	88.1	0.359	0.350	0.060
	Muon-style	86.9	0.137	0.241	0.016
	GNG-Muon	87.3	0.137	0.241	0.015
CIFAR-10	AdamW	74.6	0.367	0.338	0.088
	SGD/Nest.	71.5	0.896	0.900	0.679
	Lion	74.2	0.400	0.336	0.188
	Sophia	74.7	0.345	0.335	0.035
	Muon-style	74.8	0.228	0.263	0.018
	GNG-Muon	73.2	0.199	0.252	0.020

The long audits record 4096 pre-step gradients and updates for three seeds on MNIST/Fashion-MNIST and four on CIFAR-10. Bounded diagonal entries lie in $[10^{-6}, 1]$; layer scalar uses one positive scalar per tensor.

A follow-up campaign adds channel scalar, an Adafactor-style row, paired protocol changes, and a tiny-ResNet. The intervention table reports ablated-minus-base paired means and sample standard deviations across three seeds.

The bundled SGD/Nesterov and Lion sign/scale interventions change both accuracy and geometric residuals; AdamW weight-decay removal has little effect over the short trace. These pairs estimate bundled contrasts and do not isolate momentum from Nesterov lookahead or sign from update scale. They also lack factorial masks and hidden-state measurements. Accordingly, they support response-class relevance in nonconvex training, while the digits experiment provides the direct causal-geometric closure. The current experiments validate the P1 reduced-value law and the P3 identification/inference layer; the more general observable-readout law in Theorem 3 remains an exact analytical prediction for a future nonconvex factorial campaign.

9 Discussion, Boundaries, and Conclusion

What is complete. Pathwise history realization is complete for finite-horizon innovation-driven laws under the fixed coupling, Möbius inversion is complete for lower-finite intervention responses, and the observation kernel is the complete indistinguishability gauge for a declared finite support. Completeness concerns representation and identification under a fixed protocol; it does not provide a distribution-only minimal stochastic state or a universal predictive transition map across protocols.

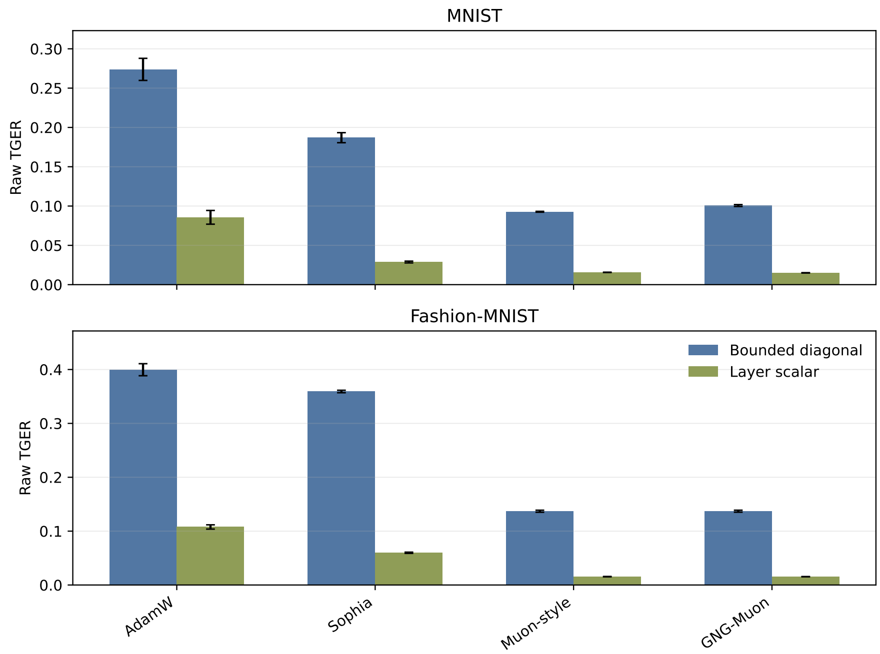

Figure 3: Response-class sensitivity. Layer scalar is nested inside bounded diagonal geometry, and the explanation-rate drop varies by optimizer.

Table 4: Paired protocol interventions. The SGD contrast jointly removes momentum and Nesterov structure; the Lion contrast removes the sign operator and changes update scale at fixed learning rate.

Intervention	Pair	ΔAcc.	ΔBDiag	ΔChannel	ΔLayer
weight decay	AdamW $\to$ AdamW/no WD	-0.2±0.1	+0.000±0.001	-0.000±0.001	+0.000±0.000
momentum/Nesterov	SGD/Nest. $\to$ SGD/no mom.	-18.9±0.1	+0.131±0.009	+0.218±0.004	+0.255±0.005
sign/scale	Lion $\to$ Lion/no sign	-39.5±1.9	+0.191±0.049	+0.052±0.036	+0.071±0.036

What the interaction layer adds. P1 explains reduced-value interaction through inverse vertical stiffness, and P2 gives the path-space version. The new readout theorem shows how that hidden response reaches an experimentally selected update, trace, or task observable. Its extra readout and second-response terms explain why an actual optimizer response need not inherit the P1 negative-semidefinite sign. The same observed pair effect can admit multiple hidden models, so the calculus does not identify geometry from responses alone. Nonunique, nonsmooth, or hysteretic hidden responses require generalized derivatives or set-valued effects.

Projection and causality. A small residual proves proximity to a response class. It does not allocate causal credit. A passive top configuration leaves $d(|\mathcal{K}|-1)$ supported coordinates ambiguous. Controlled configurations, additional instrumentation, or structural restrictions are required for attribution.

The price of exact structure. Unrestricted Boolean attribution requires 2^m configurations, and its saturated Gaussian risk grows as $(3+2\sqrt{2})^m/N$. Scientific tractability comes from falsifiable structure: low interaction order, sparse hierarchy, graded interventions, or a target decision subspace. Adaptive support selection requires selective-inference or multiplicity control beyond the fixed-support theorem.

Statistical boundary. Exact minimax, confidence, and chi-square conclusions assume a finite-dimensional linear experiment with known positive-definite covariance. Estimated covariance, dependent traces, and adaptive designs need additional analysis. Randomization calibration is exact only under a defensible invariant group action.

Table 5: CIFAR-10 small-ConvNet geometry ladder. The three columns are nested response families, not optimizer rankings.

Opt.	Acc. (%)	BDiag	Channel	Layer
AdamW	61.6	0.197	0.109	0.072
AdamW/no WD	61.4	0.198	0.109	0.072
SGD/Nest.	48.7	0.750	0.510	0.471
SGD/no mom.	29.8	0.881	0.728	0.727
Lion	50.3	0.412	0.241	0.197
Lion/no sign	10.8	0.603	0.293	0.268
Sophia	45.8	0.284	0.072	0.041
Adafactor-style	46.2	0.179	0.173	0.162
Muon-style	72.8	0.255	0.028	0.015
GNG-Muon	72.3	0.229	0.025	0.013

Table 6: CIFAR-10 tiny-ResNet geometry ladder under the same short-horizon response audit.

Opt.	Acc. (%)	BDiag	Channel	Layer
AdamW	58.6	0.300	0.159	0.124
SGD/Nest.	46.7	0.515	0.342	0.273
Lion	47.9	0.405	0.173	0.145
Sophia	39.5	0.392	0.099	0.071
Adafactor-style	40.7	0.217	0.216	0.210
Muon-style	61.0	0.173	0.043	0.027

Empirical boundary. The digits experiment directly validates the P1 reduced-value curvature law and the P3 incidence, design, inference, and falsification layers under global strong convexity and additive interventions. It does not validate the general readout-transfer law for an actual nonconvex optimizer trace. The neural experiments establish response-class sensitivity under fixed protocols, but they lack factorial masks and hidden-response measurements. Broad optimizer superiority would additionally require matched hyperparameter search, compute, throughput, memory, and modern workloads.

Research direction. The immediate theoretical target is to specialize the readout-transfer law to the full SPD and path-space realizations supplied by the companion papers. The immediate empirical target is a nonconvex geometry–memory–operator campaign that measures update readouts, complete factorial masks, and the hidden first and second responses required by equation (13).

Conclusion. Causal optimizer interaction calculus separates what an optimizer can generate, what an intervention changes, and what an experiment identifies. The universal layer covers innovation-driven finite-horizon rules under a fixed coupling. P1 and P2 supply the hidden Schur structure; the new readout layer transports it to actual optimizer observables, and the incidence layer states exactly which resulting effects an experiment can recover. This organization turns the four-paper geometry program into a concrete experimental theory without duplicating the companion structural theorems or forcing general optimizers into a gradient-flow model.

10 Reproducibility Statement

All deterministic claims are proved in the appendices. The universal causal and incidence theorem is proved in section A; Appendix B reproduces the attributed P1 reduction proof for self-containment and proves the new observable-readout transfer; the observation/inference theorem is proved in section C; and the geometric expressivity results are proved in section D.

The deterministic verification companion is `experiments/optimizer_calculus/verify_calibrated_calculus.py`. It checks incidence inversion, design rank, quotient algebra, Gaussian risk and coverage, sector tests, misspecification, randomization calibration, and action certificates. The complete factorial closure is implemented by `experiments/optimizer_calculus/interaction_curvature_experiment.py` with `experiments/optimizer_calculus/interaction_curvature_config.json`; six derivative, curvature, design, inference, and finite-response tests are in `experiments/optimizer_calculus/test_interaction_curvature_experiment.py`, and `experiments/optimizer_calculus/verify_interaction_curvature_results.py` independently audits the saved CSV/JSON artifacts. The submission includes the configuration snapshot, all summary tables, 4500 confirmatory minibatch observations, and result JSON under `newpapers/03-geometric-nongeometric-optimizer-calculus/artifacts/interaction_curvature`. Quadratic diagnostics use `experiments/toy/gng_optimizer_benchmark.py`. Trace audits use the scripts under `experiments/lane_audit`.

All random sources, quadrature orders, budgets, bootstrap draws, and power targets for the factorial closure are fixed in the configuration. The production outputs pass the independent artifact audit; the six new tests and the original calibrated-calculus verification pass. Exact software-package versions are not locked, so bitwise reproduction across numerical-library versions is not claimed.

The long runs are `paper03_rich_family_b1024_l28_20260710_2223` and `paper03_rich_cifar10_b1024_l28_20260710_2223`. They use batch size 1024, lane count 28, and 4096 recorded steps per job. MNIST/Fashion-MNIST uses seeds $7, 8, 9$; CIFAR-10 uses $7, 8, 9, 11$. Follow-up runs are `paper03_followup_sector_cifar_b1024_l28_20260711_0242` and `paper03_followup_tinyresnet_cifar_b1024_l28_20260711_0242`, using seeds $7, 8, 9$ and 128 or 96 recorded steps. Analysis commands, recorded parameters, job counts, model widths, splits, and monitor statistics are in section E. The surviving run artifacts do not bind exact Python, NumPy, PyTorch, and CUDA versions to these Paper03 jobs.

The neural scripts fix Python, NumPy, PyTorch, and data-loader seeds, but their archived configuration does not assert deterministic CUDA algorithms across hardware. The follow-up raw run directories and per-seed summaries are present in the research workspace; the long-run raw trace archives are absent from the submission, which retains their aggregate CSV/JSON summaries. The run manifest records the launch argument vector but contains `git_error` in place of a commit hash. There is no environment lockfile or complete hyperparameter-search log. Consequently the controlled factorial closure is independently auditable from supplied artifacts, while exact bitwise regeneration of every legacy neural trace is not guaranteed.

11 Ethics Statement

This work studies optimizer theory and diagnostic experiments and uses no human-subject data. The main risk is overstating causal attribution or optimizer performance from passive or incomplete interventions. We separate controlled factorial effects and support tests from passive response-class exclusion and from still-unmeasured joint effects in the neural audits.

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A Proof of the Universal Representation Theorem

We prove Theorem 1 in full.

A.1 Canonical causal realization

Fix an intervention configuration a. At time k, let h_k denote the complete extended history of oracle observations, exogenous inputs, realized innovations, and previous outputs. All component spaces are standard Borel. Causality means that, after exposing the protocol's innovation, the current response is a measurable function of h_k, the current declared input, and that innovation, without dependence on future inputs or innovations. Taking $x_k^a = h_k$ and letting Φ_k^a append the next declared input, observation, innovation, and output gives a pathwise unifilar history-state realization of equation (1). Thus existence requires no finite-memory assumption.

For extended histories at the same time, define $h \sim h'$ when every common future sequence of declared inputs and innovation values produces identical future response sequences. This is an equivalence relation. Let $[h]$ be the equivalence class. If $h \sim h'$, append the same next input and innovation value. Every later input–innovation sequence is then a common continuation of the original histories, so the two extended histories remain equivalent. The transition $[h] \mapsto [h^+]$ is therefore well-defined, and the response map is constant on equivalence classes. The classes form a sufficient pathwise predictive-state realization in the category defined in section 3.

Now let $s(h)$ be the state reached by any other admissible sufficient pathwise/unifilar realization. If $s(h) = s(h')$, its response map and unifilar updates produce the same future response sequence under every common input–innovation continuation, so $h \sim h'$. Hence

$$\pi : s(h) \longmapsto [h]$$

is a well-defined surjection from its reachable states onto the predictive quotient. A realization with two reachable states mapped to the same class is not behaviorally minimal. Every behaviorally minimal reachable realization therefore makes π bijective. Its transition and response maps are conjugate to the quotient maps, proving uniqueness up to state relabeling. This is a set-level behavioral quotient relative to the fixed innovation representation. A standard-Borel structure on the quotient requires the induced equivalence relation to be smooth; the theorem does not assume that additional descriptive-set-theoretic property.

A.2 Incidence-algebra completeness

Let $\mu_{\mathcal{P}}$ be the Möbius function of the lower-finite poset. Every sum below is finite. Define

$$\zeta_t = \sum_{s \preceq t} \mu_{\mathcal{P}}(s,t) F(s).$$

For fixed $a \in \mathcal{P}$, exchange the finite sums:

$$\begin{aligned} \sum_{t \preceq a} \zeta_t &= \sum_{t \preceq a} \sum_{s \preceq t} \mu_{\mathcal{P}}(s,t) F(s) \\ &= \sum_{s \preceq a} F(s) \sum_{t : s \preceq t \preceq a} \mu_{\mathcal{P}}(s,t). \end{aligned}$$

By the defining inverse identity for the Möbius function, the inner sum is one when $s = a$ and zero otherwise. The result is $F(a)$. Applying the same Möbius transform to any proposed reconstruction proves uniqueness.

For the Boolean lattice, the interval $[B,T]$ is a Boolean lattice of rank $|T| - |B|$, so $\mu(B,T) = (-1)^{|T|-|B|}$ and the formula reduces to mixed finite differences. For a product of chains, the product Möbius function gives the corresponding graded mixed differences.

A.3 The complete experimental gauge

Assume $\mathbb{H} \cong \mathbb{R}^d$ and stack the supported effect vectors. For queried configurations $\mathcal{D} = (a_1, \ldots, a_n)$,

$$y_{\mathcal{D}} = (X_{\mathcal{D},\mathcal{K}} \otimes I_d)\zeta, \qquad X_{\mathcal{D},\mathcal{K}}(i,t) = \mathbf{1}\{t \preceq a_i\}. \tag{30}$$

Two collections produce the same observations exactly when their difference lies in the kernel. Since

$$\operatorname{rank}(X_{\mathcal{D},\mathcal{K}} \otimes I_d) = d \operatorname{rank} X_{\mathcal{D},\mathcal{K}},$$

rank–nullity gives compatible-fiber dimension $d(|\mathcal{K}| - r_{\mathcal{D}})$ and proves the identification criterion.

If a finite poset has a greatest element $\top$, then $t \preceq \top$ for every $t \in \mathcal{K}$. The single passive row is all ones, has rank one, and leaves ambiguity dimension $d(|\mathcal{K}| - 1)$.

A.4 Sharp quotient stability

Put $A = X \otimes I_d$ and first suppose X has positive rank. Let $\widehat{\zeta} = A^\dagger y$. Since $A^\dagger A$ projects orthogonally onto $\ker A^\perp$,

$$\widehat{\zeta} - \zeta_{\mathrm{can}} = A^\dagger e = A^\dagger \operatorname{Proj}_{\operatorname{range} A} e.$$

The positive singular values of $X \otimes I_d$ are those of X, each repeated d times. Therefore

$$\left\| \widehat{\zeta} - \zeta_{\mathrm{can}} \right\| \le \frac{\left\| \operatorname{Proj}_{\operatorname{range} A} e \right\|}{\sigma^+_{\min}(X)}.$$

Equality holds when the projected error is a left singular vector associated with $\sigma^{+}_{\min}(X)$. If X has rank zero, $\ker A$ is the entire coordinate space and its canonical identifiable quotient is $\{0\}$, so there is no positive singular direction to estimate.

A.5 Exact fixed and adaptive intervention complexity

An $n \times |\mathcal{K}|$ matrix cannot have full column rank for $n < |\mathcal{K}|$, so every fixed exact design needs at least $|\mathcal{K}|$ queries. To attain the bound, query the configurations in $\mathcal{K}$. Choose a linear extension of the partial order and use it for both rows and columns. In the square matrix

$$X_{a,t} = \mathbf{1}\{t \preceq a\}, \qquad a, t \in \mathcal{K},$$

a nonzero entry can occur only when the column precedes the row, and every diagonal entry is one. The matrix is lower triangular with unit diagonal and is invertible.

For the deterministic adaptive lower bound, run a strategy with fewer than $|\mathcal{K}|$ queries against the zero effect law. Let $\mathcal{D}_0$ be its realized configurations. Choose nonzero $c \in \ker X_{\mathcal{D}_0,\mathcal{K}}$ and nonzero $v \in \mathbb{H}$, and set $h_t = c_t v$. The zero law and the law $\zeta' = h$ return identical responses at the first query. If their transcripts agree through one query, determinism selects the same next configuration, where the responses again agree by the kernel construction. Induction gives identical complete transcripts under distinct laws.

Now consider a randomized strategy. Although the ambient lower-finite poset need not be finite, a query produces one of at most $2^{|\mathcal{K}|}$ distinct incidence row patterns on the finite support $\mathcal{K}$. Fewer than $|\mathcal{K}|$ queries therefore produce one of finitely many row-pattern sequences. Under the zero law, at least one such sequence R_0 occurs with positive probability. Its stacked matrix has fewer than $|\mathcal{K}|$ rows, so choose nonzero c in its kernel, fix nonzero $v \in \mathbb{H}$, and set $h_t = c_t v$. On the event producing R_0, the zero law and h return identical responses at every queried pattern. With the same internal random choices, induction gives the same subsequent configurations and hence the same full transcript. The estimator returns the same value for two distinct laws on an event of positive probability, so it cannot be correct with probability one for both. Randomization cannot beat the lower bound.

A.6 Prediction and falsification

When $X_{\mathcal{D},\mathcal{K}}$ has full column rank, the fitting responses determine a unique ζ. The structural model forces

$$F_{\mathcal{K}}(b) = \sum_{\substack{t \in \mathcal{K} \\ t \preceq b}} \zeta_t \qquad \forall b \in \mathcal{P}.$$

A held-out response differing from this value cannot belong to the declared support class. This proves falsifiability and completes the theorem.

B Proof of the Reduction-Induced Interaction Theorem

For self-containment, we reproduce the P1 proof of Theorem 2 and Proposition 1, then prove the new observable readout law in Theorem 3.

B.1 Continuity and smoothness of the minimizer section

Fix $z_0 = (y_0, u_0)$ and write $\sigma_0 = \sigma_\star(z_0)$. Let $z_n \to z_0$. Evaluating each fiber objective at σ_0 shows that $E(z_n, \sigma_\star(z_n))$ is bounded above for large n. Local uniform inf-compactness therefore places the minimizers in one relatively compact set. Every subsequence has a further convergent subsequence, say $\sigma_\star(z_{n_j}) \to \bar{\sigma}$.

For any fixed σ, optimality gives

$$E(z_{n_j}, \sigma_\star(z_{n_j})) \le E(z_{n_j}, \sigma).$$

Continuity and passage to the limit yield $E(z_0, \bar{\sigma}) \le E(z_0, \sigma)$ for all σ. Uniqueness at z_0 implies $\bar{\sigma} = \sigma_0$. Every convergent subsequence has this limit, hence the full minimizer sequence converges. Thus $\sigma_\star$ is continuous.

The interior first-order condition is

$$E_\sigma(z, \sigma_\star(z)) = 0. \tag{31}$$

At every minimizer, $D_\sigma E_\sigma = H$ is invertible. The implicit-function theorem gives a local C^r critical section through each $(z, \sigma_\star(z))$. Continuity of the global minimizer section makes it coincide locally with that critical section. These local representations agree by uniqueness, so $\sigma_\star$ is globally C^r on the parameter domain.

B.2 Hidden response and effective Hessian

Differentiate equation (31) with respect to $z = (y, u)$:

$$E_{\sigma z} + H D_z \sigma_\star = 0.$$

Since H is invertible,

$$D_z \sigma_\star = -H^{-1} E_{\sigma z},$$

which proves equation (6).

For the reduced value, the chain rule and stationarity give the envelope identity

$$D_z \bar{E} = E_z + E_\sigma D_z \sigma_\star = E_z.$$

Differentiating again,

$$D^2_{zz} \bar{E} = E_{zz} + E_{z\sigma} D_z \sigma_\star = E_{zz} - E_{z\sigma} H^{-1} E_{\sigma z}.$$

This is equation (7). At a critical hidden point the displayed quadratic form is the Schur complement of the vertical Hessian, hence is invariant under smooth changes of hidden coordinates.

B.3 Affine interventions and interaction curvature

Under equation (8), $E_{uu} = 0$ and $E_{\sigma u} = G$. The u block of equation (7) is therefore

$$D^2_{uu} \bar{E} = -G^* H^{-1} G.$$

For every $a \in \mathbb{R}^m$,

$$a^\top D^2_{uu} \bar{E}\, a = -\left\langle Ga, H^{-1} Ga \right\rangle \le 0,$$

because H^{-1} is positive definite. This proves equation (9). Componentwise,

$$\partial^2_{u_i u_j} \bar{E} = -\left\langle D_\sigma \Delta_i, H^{-1} D_\sigma \Delta_j \right\rangle. \tag{32}$$

The same concavity conclusion holds without differentiability whenever the infimum is finite: for fixed (y, σ), the unreduced energy is affine in u, and a pointwise infimum of affine functions is concave.

B.4 Observable readout interaction

Fix y and suppress it from the notation. Write $v_i = \partial_{u_i} \sigma_\star$. Differentiating stationarity once gives

$$H v_i + E_{\sigma u_i} = 0, \qquad v_i = -H^{-1} E_{\sigma u_i}.$$

Differentiate this identity with respect to u_j along the minimizing section. The total derivative of $H = E_{\sigma\sigma}$ contributes $E_{\sigma\sigma\sigma}[v_j, v_i] + E_{\sigma\sigma u_j} v_i$, while the total derivative of $E_{\sigma u_i}$ contributes $E_{\sigma\sigma u_i} v_j + E_{\sigma u_i u_j}$. Therefore

$$H \partial^2_{u_i u_j} \sigma_\star + E_{\sigma\sigma\sigma}[v_i, v_j] + E_{\sigma\sigma u_i} v_j + E_{\sigma\sigma u_j} v_i + E_{\sigma u_i u_j} = 0.$$

Invertibility of H proves equation (12).

For $F_R(u) = R(u, \sigma_\star(u))$, the first derivative is

$$\partial_{u_i} F_R = R_{u_i} + R_\sigma v_i.$$

Differentiating with respect to u_j and applying the product and chain rules gives

$$\partial^2_{u_i u_j} F_R = R_{u_i u_j} + R_{u_i \sigma} v_j + R_{\sigma u_j} v_i + R_{\sigma\sigma}[v_i, v_j] + R_\sigma \partial^2_{u_i u_j} \sigma_\star,$$

which is equation (13). Applying the two coordinate finite-difference operators and the fundamental theorem of calculus proves equation (14). If E is affine in u, direct differentiation gives $E_{\sigma u_i u_j} = 0$ and $E_{\sigma\sigma u_i} = D^2_{\sigma\sigma}\Delta_i$. No sign follows for the readout Hessian because its five terms need not form a negative Gram matrix.

B.5 Möbius effects as integrated derivatives

For a coordinate i, let T_i replace $u_i = 0$ by $u_i = 1$. The fundamental theorem of calculus gives

$$(\mathsf{T}_i - I)\bar{E}(0) = \int_0^1 \partial_{u_i} \bar{E}(t_i e_i)\, \mathrm{d}t_i.$$

The coordinate difference operators commute. Repeating the identity for every $i \in T$ gives

$$\prod_{i\in T} (\mathsf{T}_i - I)\bar{E}(0) = \int_{[0,1]^T} \partial_T^{|T|} \bar{E}\left(\sum_{i\in T} t_i e_i\right) \prod_{i\in T} \mathrm{d}t_i.$$

Expanding the product on the left yields

$$\sum_{B\subseteq T} (-1)^{|T|-|B|} \bar{E}(\mathbf{1}_B) = \zeta_T,$$

proving equation (10). For $T = \{i, j\}$, substitution of equation (32) proves equation (11).

A zero integral does not force the integrand to vanish because mixed curvature may change sign. Pointwise vanishing is equivalent to inverse-Hessian orthogonality of $D_\sigma \Delta_i$ and $D_\sigma \Delta_j$.

B.6 Noncommutation and analytical examples

For Proposition 1, let

$$E_0(\sigma) = \sigma^2, \qquad E_1(\sigma) = (\sigma - 1)^2.$$

Both reduced values are zero, so their reduced first effect is zero. The pointwise first difference is $E_1 - E_0 = 1 - 2\sigma$, whose infimum is $-\infty$. Thus no general exchange law exists.

If instead $E_A(\sigma) = E_0(\sigma) + \sum_{i\in A} c_i$ with fiber constants c_i, then the constants leave the minimizer unchanged and move outside the infimum. The reduced response is additive, with singleton effects c_i and all higher effects zero.

Finally, for

$$E(\sigma, u) = \frac{h}{2}\sigma^2 + u_1 a\sigma + u_2 b\sigma,$$

stationarity yields $\sigma_\star = -(au_1 + bu_2)/h$. Substitution gives

$$\bar{E}(u) = -\frac{(au_1 + bu_2)^2}{2h}.$$

The Boolean pair contrast is

$$\bar{E}(1,1) - \bar{E}(1,0) - \bar{E}(0,1) + \bar{E}(0,0) = -\frac{ab}{h},$$

matching equation (11).

For the coupled log-preconditioner in equation (15), $h > |\rho|$ makes

$$H = \begin{pmatrix} h & \rho \\ \rho & h \end{pmatrix} \quad \text{positive definite,} \qquad H^{-1} = \frac{1}{h^2 - \rho^2} \begin{pmatrix} h & -\rho \\ -\rho & h \end{pmatrix}.$$

Stationarity gives

$$H\sigma - b(g) + u = 0, \qquad \sigma_\star(g,u) = H^{-1}(b(g) - u).$$

Completing the square yields the reduced scalar calibration response

$$\bar{E}(g,u) = -\frac{1}{2}(b(g) - u)^\top H^{-1}(b(g) - u).$$

Because $D_\sigma \Delta_1 = e_1$, $D_\sigma \Delta_2 = e_2$, and $G = I_2$, its mixed derivative is constant:

$$\partial^2_{u_1 u_2} \bar{E} = -e_1^\top H^{-1} e_2 = \frac{\rho}{h^2 - \rho^2}.$$

The integral over the Boolean intervention square has the same value, proving equation (16). The selected update $q(\sigma_\star, g) = -\operatorname{diag}(e^{-\sigma_{\star,1}/2}, e^{-\sigma_{\star,2}/2})g$ is a positive diagonal preconditioned gradient step for every intervention amplitude.

Let $D = h^2 - \rho^2$ and $\sigma_\star(0) = H^{-1}b(g)$. The first hidden coordinate is

$$\sigma_{\star,1}(u) = \sigma_{\star,1}(0) - \frac{h}{D}u_1 + \frac{\rho}{D}u_2.$$

Substitution into $q_1 = -g_1 e^{-\sigma_{\star,1}/2}$ yields the exponential readout displayed in the main text. For a function $q_1(0)e^{au_1 + cu_2}$, the Boolean pair difference is exactly $q_1(0)(e^a - 1)(e^c - 1)$. Taking $a = h/(2D)$ and $c = -\rho/(2D)$ proves equation (17) and completes the optimizer realization.

C Proof of the Projection–Information–Decision Theorem

C.1 Projection, duality, stability, and nesting

Because $\mathcal{M}$ is nonempty, closed, and convex in finite dimension, the strictly convex coercive function $m \mapsto \frac{1}{2}\|y - m\|^2_{V^{-1}}$ has a unique minimizer $\widehat{m}$. Its first-order variational inequality is

$$(y - \widehat{m})^\top V^{-1}(m - \widehat{m}) \le 0 \qquad \forall m \in \mathcal{M}. \tag{33}$$

For arbitrary q and m, weighted Young's inequality gives

$$q^\top(y - m) - \frac{1}{2}q^\top V q \le \frac{1}{2}\|y - m\|^2_{V^{-1}}.$$

Taking the infimum over $m \in \mathcal{M}$ yields

$$q^\top y - \sigma_{\mathcal{M}}(q) - \frac{1}{2}q^\top V q \le \frac{1}{2}\delta_{\mathcal{M}}(y)^2.$$

Set $q^\star = V^{-1}(y - \widehat{m})$. By equation (33), $(q^\star)^\top m \le (q^\star)^\top \widehat{m}$ for all $m \in \mathcal{M}$; hence $\sigma_{\mathcal{M}}(q^\star) = (q^\star)^\top \widehat{m}$. Substitution gives equality and proves equation (18).

The distance stability follows from the triangle inequality. For any $m \in \mathcal{M}$,

$$\delta_{\mathcal{M}}(y) \le \|y - y'\|_{V^{-1}} + \|y' - m\|_{V^{-1}}.$$

Taking the infimum over m and exchanging y, y' proves the one-Lipschitz bound. If $\mathcal{M}_1 \subseteq \mathcal{M}_2$, the infimum defining distance is taken over a larger set for $\mathcal{M}_2$, proving nesting monotonicity.

C.2 Observational quotient and exact minimax risk

In the model $Y = Az + \varepsilon$ with common covariance V, two Gaussian laws are equal exactly when their means agree. Therefore

$$P_z = P_{z'} \iff A(z - z') = 0 \iff z - z' \in K.$$

Every coset has one orthogonal representative in $K^{\perp}$.

For $z \in K^{\perp}$, $\mathcal{I}^{\dagger}\mathcal{I}z = z$, and

$$\widehat{z} - z = \mathcal{I}^{\dagger} A^{*} V^{-1} \varepsilon.$$

Its covariance is

$$\begin{aligned}\operatorname{Cov}(\widehat{z} - z) &= \mathcal{I}^{\dagger} A^{*} V^{-1} V V^{-1} A \mathcal{I}^{\dagger} \\ &= \mathcal{I}^{\dagger}\mathcal{I}\mathcal{I}^{\dagger} = \mathcal{I}^{\dagger}.\end{aligned}$$

The risk is therefore $\operatorname{tr}(\mathcal{I}^{\dagger})$ for every canonical z.

For the matching lower bound, restrict the experiment to $K^{\perp}$, where $\mathcal{I}$ is positive definite, and use the Gaussian prior $z \sim N(0, \tau^2 I)$. The posterior covariance is

$$C_{\tau} = (\tau^{-2} I + \mathcal{I})^{-1}.$$

The Bayes risk of the posterior mean is $\operatorname{tr}(C_{\tau})$ and lower-bounds the minimax risk. Letting $\tau \to \infty$ in the eigenvalue formula gives

$$R_{\text{minimax}} \geq \operatorname{tr}(\mathcal{I}^{-1}|_{K^{\perp}}) = \operatorname{tr}(\mathcal{I}^{\dagger}).$$

Generalized least squares attains the bound.

Let $\eta = V^{-1/2}\varepsilon \sim N(0, I)$. The whitened fitted-mean error is

$$V^{-1/2} A(\widehat{z} - z) = \operatorname{Proj}_{\operatorname{range}(V^{-1/2}A)} \eta.$$

Its squared norm is $(\widehat{z} - z)^{*}\mathcal{I}(\widehat{z} - z)$ and the projection has rank r, proving the exact χ^2_r confidence law.

C.3 Residualized block tests

Partition $A = [A_{-G} \ A_G]$ and set

$$\widetilde{A} = V^{-1/2} A, \qquad \Pi_{-G} = \operatorname{Proj}_{\operatorname{range}(\widetilde{A}_{-G})}, \qquad B_G = (I - \Pi_{-G})\widetilde{A}_G.$$

The range of B_G is orthogonal to the nuisance range. For $\widetilde{Y} = V^{-1/2} Y$,

$$\operatorname{Proj}_{\operatorname{range}(B_G)} \widetilde{Y} = B_G z_G + \operatorname{Proj}_{\operatorname{range}(B_G)} \eta.$$

Under $B_G z_G = 0$, the squared norm has a central chi-square law with rank B_G degrees of freedom. Under the alternative it has a noncentral chi-square law with noncentrality $\|B_G z_G\|^2$.

C.4 Exact misspecification decomposition

Suppose the true mean is $Az + b$ with $z \in K^{\perp}$. Substitution into GLS gives

$$\widehat{z} - z = \mathcal{I}^{\dagger} A^{*} V^{-1} b + \mathcal{I}^{\dagger} A^{*} V^{-1} \varepsilon = b_{\parallel} + \text{zero-mean noise}.$$

The noise covariance remains $\mathcal{I}^{\dagger}$, so the cross term vanishes after expectation and

$$\mathbb{E}\,\|\widehat{z} - z\|^2 = \left\|b_{\parallel}\right\|^2 + \operatorname{tr}(\mathcal{I}^{\dagger}).$$

Let Π_A project onto $\operatorname{range}(V^{-1/2}A)$. Then

$$(I - \Pi_A)V^{-1/2} Y = (I - \Pi_A)V^{-1/2} b + (I - \Pi_A)\eta.$$

The projection has dimension $n - r$, and the deterministic shift has squared norm $\|b_{\perp}\|^2$. The displayed lack-of-fit statistic is therefore $\chi^2_{n-r}(\|b_{\perp}\|^2)$.

C.5 Transfer and action selection

Let $r_y = y - \widehat{m}$. Since $r_y = V^{1/2}(V^{-1/2} r_y)$,

$$\|L r_y\| \leq \left\|L V^{1/2}\right\|_{\mathrm{op}} \left\|V^{-1/2} r_y\right\| = \left\|L V^{1/2}\right\|_{\mathrm{op}} \delta_{\mathcal{M}}(y).$$

On the exact confidence event,

$$(\widehat{z} - z)^* \mathcal{I} (\widehat{z} - z) \leq \chi^2_{r,1-\delta}.$$

For $\ell_\pi \in K^\perp = \operatorname{range} \mathcal{I}$, generalized Cauchy–Schwarz yields, simultaneously for every declared action,

$$|\ell_\pi^\top (\widehat{z} - z)| \leq \sqrt{\chi^2_{r,1-\delta}} \sqrt{\ell_\pi^\top \mathcal{I}^\dagger \ell_\pi} = \rho_\pi(\delta).$$

Hence

$$\begin{aligned} \Delta(\widehat{\pi}; z) &\leq \Delta(\widehat{\pi}; \widehat{z}) + \rho_{\widehat{\pi}} \\ &\leq \Delta(\pi^\star; \widehat{z}) + \rho_{\pi^\star} \\ &\leq \Delta(\pi^\star; z) + 2\rho_{\pi^\star}. \end{aligned}$$

C.6 Exact optimal replication

For the saturated design, X is square and invertible. With $W = \operatorname{diag}(n_a)$ and $M = X^{-1}$,

$$\mathcal{I}^{-1} = (X^{-1} W^{-1} X^{-\top}) \otimes \Sigma = (M W^{-1} M^\top) \otimes \Sigma.$$

Therefore

$$\operatorname{tr}(\mathcal{I}^{-1}) = \operatorname{tr}(\Sigma) \sum_a \frac{\|M_{:,a}\|_2^2}{n_a}.$$

Writing $c_a = \|M_{:,a}\|_2^2$, Cauchy–Schwarz gives

$$\left(\sum_a \sqrt{c_a}\right)^2 = \left(\sum_a \sqrt{\frac{c_a}{n_a}} \sqrt{n_a}\right)^2 \leq \left(\sum_a \frac{c_a}{n_a}\right) N.$$

Equality holds exactly when n_a is proportional to $\sqrt{c_a}$, proving equations (22) and (23).

For the full Boolean lattice, Möbius inversion gives

$$M_{T,A} = (-1)^{|T|-|A|} \mathbf{1}\{A \subseteq T\}.$$

Column A has $2^{m-|A|}$ nonzero entries of squared magnitude one, so $c_A = 2^{m-|A|}$. Moreover,

$$\sum_{A \subseteq [m]} \sqrt{c_A} = \sum_{j=0}^{m} \binom{m}{j} 2^{(m-j)/2} = (1 + \sqrt{2})^m.$$

Since $(1+\sqrt{2})^{2m} = (3 + 2\sqrt{2})^m$, substitution proves equation (24). This completes the proof of Theorem 4.

C.7 Target-optimal replication

For scalar independent configuration means, the saturated effect estimator is $\widehat{\zeta} = M\bar{Y}$ and

$$\operatorname{Cov}(L\widehat{\zeta}) = LM \operatorname{diag}\left(\frac{\sigma_a^2}{n_a}\right) M^\top L^\top.$$

Taking the trace and expanding by columns gives

$$R_L(n) = \sum_a \frac{\sigma_a^2}{n_a} \|LM_{:,a}\|_2^2 = \sum_a \frac{c_a \sigma_a^2}{n_a}.$$

Cauchy–Schwarz yields

$$\left(\sum_a \sigma_a \sqrt{c_a}\right)^2 = \left(\sum_a \frac{\sigma_a \sqrt{c_a}}{\sqrt{n_a}} \sqrt{n_a}\right)^2 \leq R_L(n) N.$$

Equality holds exactly when $n_a \propto \sigma_a \sqrt{c_a}$, proving equations (25) and (26).

C.8 Exact group-randomization calibration

Let G be finite and let the null law of Y be invariant under every $g \in G$. Conditional on an orbit, invariance makes the realized orbit point uniform with the multiplicities induced by the group action. Among the $|G|$ transformed statistics, at most an α fraction can have an upper rank whose normalized count is at most α. Therefore

$$\Pr_0\{p_G(Y) \leq \alpha \mid \mathrm{Orb}(Y)\} \leq \alpha.$$

Averaging over orbits proves unconditional super-uniformity. Ties make the nonrandomized upper-tail value conservative.

C.9 Analytic diagonal sign null

For unbounded positive diagonal geometry, the coordinate residual is zero when $u_i g_i < 0$ and has squared magnitude u_i^2 when $u_i g_i > 0$; zero-update coordinates carry no weight. Conditional on fixed nonzero magnitudes and independent fair gradient signs,

$$R = \frac{\rho_{\mathrm{diag,I}}(u;g)^2}{\|u\|^2} = \sum_i w_i B_i, \qquad w_i = \frac{u_i^2}{\sum_j u_j^2}, \qquad B_i \overset{\mathrm{iid}}{\sim} \mathrm{Bernoulli}(1/2).$$

Independence gives

$$\mathbb{E}R = \frac{1}{2}, \qquad \mathrm{Var}(R) = \frac{1}{4}\sum_i w_i^2.$$

Weighted Hoeffding gives

$$\Pr\{|R - 1/2| \geq t\} \leq 2\exp\left(-\frac{2t^2}{\sum_i w_i^2}\right) = 2e^{-2d_{\mathrm{eff}} t^2}.$$

If $d_{\mathrm{eff}} \to \infty$, then $R \to 1/2$ in probability. Continuity of $x \mapsto 1 - \sqrt{x}$ gives TGER convergence to $1 - 1/\sqrt{2}$.

D Proofs

D.1 Proof of Proposition 2

If $u = -Pg$ with $P \succ 0$ and $g \neq 0$, then

$$g^\top u = -g^\top P g < 0.$$

Conversely, suppose $g^\top u < 0$ and define $b = -u$. Then $g^\top b > 0$. Choose an orthonormal basis with $e_1 = g/\|g\|$. In this basis write

$$b = \beta e_1 + c, \qquad c \perp e_1, \qquad \beta = e_1^\top b = \frac{g^\top b}{\|g\|} > 0.$$

For $d = 1$, set $P = [\beta / \|g\|]$. For $d > 1$, define

$$\widetilde{P} = \begin{pmatrix} \beta / \|g\| & c^\top / \|g\| \\ c / \|g\| & \gamma I \end{pmatrix}$$

in the chosen basis. Then

$$\widetilde{P}(\|g\| \, e_1) = b.$$

The Schur complement of the upper-left block is

$$\gamma I - \frac{cc^\top}{\|g\| \; \beta}.$$

Choosing $\gamma > \|c\|^2 / (\|g\| \; \beta)$ makes this complement positive, so $\widetilde{P} \succ 0$. Transforming back to the original coordinates gives an SPD matrix P with $Pg = b = -u$, hence $u = -Pg$.

D.2 Proof of Corollary 2

If $\mathrm{d}f_\theta = 0$, linearity gives $P_\theta \mathrm{d}f_\theta = 0$ for every positive cometric P_θ. Thus a pure positive-geometry update is zero. Any nonzero tangent vector at that point must enter through another module or through a different modeling convention.

D.3 Proof of Proposition 3

The equation $u = -Pg$ with $P = \mathrm{diag}(p_1, \ldots, p_d)$ and $p_i > 0$ is equivalent coordinate-wise to

$$u_i = -p_i g_i.$$

If $g_i = 0$, exact expression requires $u_i = 0$. If $g_i \neq 0$, exact expression requires $p_i = -u_i/g_i > 0$, equivalently $u_i g_i < 0$.

For the residual, the squared Euclidean objective separates:

$$\inf_{p_i > 0} \sum_i (u_i + p_i g_i)^2 = \sum_i \inf_{p_i > 0} (u_i + p_i g_i)^2.$$

If $g_i = 0$, the term is u_i^2. If $g_i \neq 0$ and $u_i g_i < 0$, the positive choice $p_i = -u_i/g_i$ gives zero. If $g_i \neq 0$ and $u_i g_i > 0$, the unconstrained minimizer is negative, so the infimum over $p_i > 0$ is the boundary value u_i^2, approached as $p_i \downarrow 0$. If $g_i \neq 0$ and $u_i = 0$, the infimum is zero, again approached as $p_i \downarrow 0$, but it is not attained by a strictly positive p_i. This proves the formula and the attainment statement.

D.4 Bounded diagonal residual

Proposition 5 (Bounded diagonal residual)**.** *Fix $0 < \lambda \leq L < \infty$ and let*

$$\mathcal{P}_{\mathrm{diag}}^{[\lambda, L]} = \{\mathrm{diag}(p_1, \ldots, p_d) : \lambda \leq p_i \leq L\}.$$

For $g, u \in \mathbb{R}^d$ and $h = I$,

$$\rho_{\mathrm{diag}^{[\lambda,L]}, I}(u; g)^2 = \sum_{i=1}^{d} (u_i + p_i^\star g_i)^2,$$

where, for $g_i \neq 0$,

$$p_i^\star = \min\left\{ L, \max\left\{ \lambda, -\frac{u_i}{g_i} \right\} \right\},$$

and for $g_i = 0$ any $p_i^\star \in [\lambda, L]$ gives the same term u_i^2.

Proof. The squared residual separates by coordinate:

$$\inf_{\lambda \le p_i \le L} \sum_i (u_i + p_i g_i)^2 = \sum_i \inf_{\lambda \le p_i \le L} (u_i + p_i g_i)^2.$$

If $g_i = 0$, the coordinate objective is constant and equal to u_i^2. If $g_i \neq 0$, the unconstrained minimizer of the one-dimensional convex quadratic is $-u_i/g_i$. Projecting this minimizer onto the interval $[\lambda, L]$ gives $p_i^\star$, and substituting the coordinate minimizers gives the stated formula. □

D.5 Proof of Proposition 4

For a block-diagonal P, the equation $u = -Pg$ separates over blocks:

$$u_{B_j} = -P_{B_j} g_{B_j}.$$

If $g_{B_j} = 0$, exact expression requires $u_{B_j} = 0$. If $g_{B_j} \neq 0$, Proposition 2 applied inside the block gives existence of $P_{B_j} \succ 0$ exactly when $g_{B_j}^\top u_{B_j} < 0$. Combining the block solutions gives the desired block-diagonal matrix.

E Diagnostic Prototype Details

E.1 Controlled factorial closure

The controlled experiment is generated from existing scikit-learn digits data by

```
python -u experiments/optimizer_calculus/interaction_curvature_experiment.py ‘
  --output-dir experiments/optimizer_calculus/interaction-curvature-results
```

The complete configuration is `experiments/optimizer_calculus/interaction_curvature_config.json`. The paper-supplied mirror under `artifacts/interaction_curvature` contains the final configuration snapshot, Boolean responses, pair-curvature checks, continuous held-out checks, split robustness, allocations, Gaussian campaign summaries, all 4500 confirmatory empirical observations, bootstrap intervals, finite-response traces, and the consolidated summary.

The dataset is restricted to digits 3 and 8 and stratified into 249 training and 108 test examples. Standardization is fit on the training data only. An intercept gives 65 hidden coordinates, all ridge-regularized with $\lambda = 0.15$. The three additive losses emphasize positive-class samples, negative-class samples, and central-occlusion robustness with scale 0.75. Newton solves use analytic gradients and Hessians, Armijo damping, tolerance 10^{-11}, and at most 80 iterations. Pair curvature uses order-24 Gauss–Legendre quadrature; eight additional splits use order 16.

The Gaussian pilot allocates 420 main and 80 held-out replications. Exact noncentral-χ^2 power calibration multiplies both by the first integer reaching 0.95 power, which is nine. Confirmatory simulation uses 100,000 campaigns. The empirical channel uses minibatches of size 1024, 200 pilot observations per mask, a disjoint confirmatory sample of 3780 main and 720 held-out observations, and 20,000 stratified bootstrap draws. The pilot is used only for variance estimation and allocation.

Finite-response tracking runs fixed-step gradient descent for horizons $0, 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024$. At every horizon the script verifies that the actual pair-effect error is below the sum of the four optimized-value gaps, and that this exact gap bound is below the analytic strong-convexity bound.

The verification commands are

```
python -m pytest ‘
  experiments/optimizer_calculus/test_interaction_curvature_experiment.py -q

python experiments/optimizer_calculus/verify_interaction_curvature_results.py ‘
  newpapers/03-geometric-nongeometric-optimizer-calculus/artifacts/interaction_curvature
```

They return `6 passed` and `interaction-curvature result audit passed`, respectively.

E.2 Quadratic benchmark

The deterministic quadratic benchmark uses three problem families:

1. rotated strongly convex quadratics;
2. diagonal-plus-low-rank quadratics;
3. block-coupled quadratics.

The script records final objective gap, gradient-call count, and convergence under a gradient-norm tolerance. The default repository command is

```
python experiments/toy/gng_optimizer_benchmark.py --output experiments/toy/gng-results
```

and the larger diagnostic run used during development was

```
python experiments/toy/gng_optimizer_benchmark.py --output experiments/toy/gng-results ‘
  --dim 64 --instances 4 --memory 8 --max-iter 220 --grad-tol 1e-8
```

The paper reports the later optimized diagnostic checks because they use GNG-FullMetricProbe v2, which removes an unnecessary final gradient validation call and therefore uses $d + 1$ gradient calls (65 when $d = 64$), as recorded in `experiments/toy/gng-results-optimized-64x1/gng_optimizer_summary.json`. The saved diagnostic summary also contains HeavyBall-oracle; its poor performance on these high-condition-number instances should not be read as a general statement about heavy-ball methods. The Adam-tuned entry in that summary is selected from a six-point learning-rate sweep, whose tuning cost is separate from the recorded optimizer-call count.

E.3 Prototype descriptions

GNG-FullMetricProbe. The method evaluates $g_0 = \nabla f(x_0)$ and then evaluates gradients at $x_0 + e_i$ for $i = 1, \dots, d$. For a deterministic quadratic, these probes recover the columns of H. The update $x_+ = x_0 - H^{-1} g_0$ is the exact Newton step. The final gradient can be computed internally as $g_0 + H(x_+ - x_0)$, so no final oracle call is needed.

GNG-KrylovMetric. The method performs sequential directional probes to obtain Hp along search directions. It uses exact quadratic line search and conjugacy-style updates. Its interpretation in this paper is a growing Krylov subspace metric, not a new claim about conjugate-gradient theory. The reported implementation is not presented as a standard CG baseline.

RSDLR-BFGS-Hybrid. The method combines a bounded diagonal inverse metric with a limited secant memory. It keeps a mixture of recent secant pairs and pairs with high residual under the current diagonal model. In the present benchmark it did not outperform FIFO-LBFGS.

E.4 Formal trace audit

The trace-level residual audit in section 8 is produced by the scripts under `experiments/lane_audit`. Each worker runs `optimizer_trace.py`, records the raw gradient before the optimizer step, applies the optimizer, records the parameter update, and writes `trace_residual_summary.json`, `trace_residuals.csv`, and a GER plot. In the rich runs, each worker also writes bounded-diagonal sensitivity metrics, alternative visible-covector metrics, layer-scalar metrics, penalized layer-scalar path-fit curves, partial CSV files, progress JSON, and explicit checkpoints. The queued batch is launched by `lane_orchestrator.py`. The paper tables and figures are generated by:

```
python experiments/lane_audit/analyze_formal_run.py ‘
  --run-dir tmp/remote_runs/paper03_rich_family_b1024_l28_20260710_2223 ‘
```

```
  --output-dir newpapers/03-geometric-nongeometric-optimizer-calculus/figures `
  --prefix family_sensitivity

python experiments/lane_audit/analyze_formal_run.py `
  --run-dir tmp/remote_runs/paper03_rich_cifar10_b1024_l28_20260710_2223 `
  --output-dir newpapers/03-geometric-nongeometric-optimizer-calculus/figures `
  --prefix cifar10_audit

python experiments/lane_audit/analyze_paper03_rich_runs.py `
  --run-dir tmp/remote_runs/paper03_rich_family_b1024_l28_20260710_2223 `
            tmp/remote_runs/paper03_rich_cifar10_b1024_l28_20260710_2223 `
  --output-dir newpapers/03-geometric-nongeometric-optimizer-calculus/figures `
  --prefix paper03_rich
```

Both formal batches use lane count 28, batch size 1024, checkpoint and partial-write interval 256, and 4096 recorded optimizer steps per job. The formal jobs did not download datasets on the remote hosts. The bounded diagonal family is $A_{ii} \in [10^{-6}, 1]$ unless the sensitivity grid names a different lower or upper bound. Raw TGER is the ℓ_2 trace explanation rate when each step chooses its own map. Coherent TGER is the no-variation $B_{\text{geo}} = 0$ specialization: one shared map is fit to the whole trace. The coherent residual is computed by the closed-form streaming accumulator in `optimizer_trace.py`; it solves the same least-squares problem as stacking all updates and gradients while avoiding full trace-vector storage in host memory. In the audit script, the names `muon` and `gng_muon` both call the local GNGMuon implementation with different hyperparameter settings. They are retained as implementation labels for the saved summaries, but they should not be interpreted as official reference implementations of external Muon optimizers.

The family-sensitivity rich run is `paper03_rich_family_b1024_l28_20260710_2223`. It uses datasets `mnist,fashion_mnist`, optimizers `adamw,sophia,muon,gng_muon`, seeds `7,8,9`, width 32, train subset 8192, and test subset 2048. It contains 24 jobs, 98304 optimizer steps, and approximately 1.01×10^8 sample views. It ran on one NVIDIA RTX PRO 6000 Blackwell Server Edition GPU; the batch monitor recorded 2990 seconds of wall time, mean GPU utilization 99.5%, peak GPU utilization 100%, and peak VRAM 33015 MiB.

The CIFAR-10 rich run is `paper03_rich_cifar10_b1024_l28_20260710_2223`. It uses dataset `cifar10`, the full 50000/10000 train/test split, width 48, optimizers `adamw,sgd,lion,sophia,muon,gng_muon`, and seeds `7,8,9,11`. It contains 24 jobs, 98304 optimizer steps, and approximately 1.01×10^8 sample views. It ran on one NVIDIA RTX PRO 6000 Blackwell Server Edition GPU; the batch monitor recorded 3554 seconds of wall time, mean GPU utilization 95.7%, peak GPU utilization 100%, and peak VRAM 53373 MiB.

E.5 Follow-up trace audits

The runtime-constrained follow-up tables in tables 4 to 6 are generated by:

```
$SECTOR="tmp/remote_runs/<sector-followup-run>"
$TINY="tmp/remote_runs/<tinyresnet-followup-run>"
$OUT="newpapers/03-geometric-nongeometric-optimizer-calculus/figures"

python experiments/lane_audit/analyze_paper03_followup_runs.py `
  --sector-run-dir $SECTOR `
  --tiny-run-dir $TINY `
  --output-dir $OUT `
  --prefix paper03_followup
```

The analyzer checks that the sector run has 30 optimizer/seed rows, the tiny-ResNet run has 18 rows, the final short-horizon step counts are 128 and 96, and all reported accuracy, bounded-diagonal, channel-scalar, layer-scalar, coherent, and alternative-covector metrics are present.

The sector/intervention run is:

Table 7: Synthetic residual sanity check used before the neural-network follow-up audits. Raw TGER allows the explaining geometry to vary by step; coherent TGER uses one shared explaining map across the trace.

Mechanism	Mean GER	Raw TGER	Coh. TGER	Gap
constant diagonal	1.000	1.000	1.000	0.000
varying diagonal	1.000	1.000	0.520	0.480
momentum memory	0.484	0.455	0.227	0.227
decoupled decay	0.925	0.924	0.761	0.163
operator mixing	0.785	0.785	0.494	0.291

`paper03_followup_sector_cifar_b1024_l28_20260711_0242`

It uses CIFAR-10 with the full 50000/10000 split, width 48, the `small_conv` model, batch size 1024, lane count 28, seeds $7, 8, 9$, and ten optimizer settings: AdamW, AdamW/no weight decay, SGD/Nesterov, SGD/no momentum, Lion, Lion/no sign, Sophia, Adafactor-style, Muon-style, and GNG-Muon. Each job records 128 optimizer steps with partial trace writes and checkpoints every 32 steps. The batch contains 30 jobs and 3840 recorded optimizer steps. Its monitor recorded 2569 seconds of batch wall time, mean GPU utilization 91.4%, peak GPU utilization 100%, and peak VRAM 58757 MiB.

The tiny-ResNet follow-up run is:

`paper03_followup_tinyresnet_cifar_b1024_l28_20260711_0242`

It uses CIFAR-10 with the full 50000/10000 split, width 24, the `tiny_resnet` model, batch size 1024, lane count 28, seeds $7, 8, 9$, and six optimizer settings: AdamW, SGD/Nesterov, Lion, Sophia, Adafactor-style, and Muon-style. Each job records 96 optimizer steps with partial trace writes and checkpoints every 32 steps. The batch contains 18 jobs and 1728 recorded optimizer steps. Its monitor recorded 2482 seconds of batch wall time, mean GPU utilization 99.7%, peak GPU utilization 100%, and peak VRAM 49499 MiB.

The follow-up campaign initially tested longer horizons, but calibration with the channel-scalar and coherent covector diagnostics projected runtimes beyond the intended two-hour-per-task budget. The final reported follow-up runs therefore keep the full optimizer/seed/geometry/covector matrix and shorten the trace horizon. They are used as mechanism and model-extension audits, not as replacement performance benchmarks for the longer rich trace runs.